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Design of an Asynchronous Flash Analog-to-Digital Converter

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Abstract

The growth of the Internet of Things and of biomedical sensing has created a strong demand for low-power signal processing interfaces. In many of these applications the signals of interest are sparse: short bursts of information separated by long periods in which nothing happens. Classical synchronous analog-to-digital converters handle such signals poorly. Their global clock forces them to dissipate dynamic power at a fixed rate, whether or not the input carries any new information.

Ideally, the sampling rate would adapt to the signal itself, so that no energy is spent on clock activity during idle intervals. Such adaptation is not straightforward to implement. An asynchronous ADC is a natural way to achieve it, since conversion is driven by the signal rather than by a clock, but the architecture must be redesigned to operate correctly without a global timing reference.

In a Flash converter, the case for asynchronous control rests first on the elimination of the high-speed clock distribution network, which is a substantial fraction of the power budget. Clockless operation also simplifies the treatment of comparator metastability and shortens the conversion latency, which matters in event-driven applications where signal activity varies widely over time.

This dissertation presents the design and implementation of an asynchronous Flash ADC. With the clock removed, power consumption follows the activity of the input signal, and the dynamic component falls to near zero when the input is idle. An offline body bias trimming strategy compensates the deterministic trip-point deviations of the comparison stage across process corners without affecting the speed of the Flash topology. The work covers the theoretical analysis, the schematic design and the validation of the system through analog and mixed-signal co-simulation. The result is a robust, clockless converter with a better figure of merit than conventional synchronous architectures for sparse-signal applications.

Keywords: Analog-to-Digital Converter, Asynchronous Design, Flash ADC, Level-Crossing Sampling, Offset Trimming, Low Power, IoT.

UN Sustainable Development Goals

The United Nations Sustainable Development Goals (SDGs) provide a global framework to achieve a better and more sustainable future for all. It includes 17 goals to address the world's most pressing challenges, including poverty, inequality, climate change, environmental degradation, peace, and justice.

Electronic systems play a pivotal role in modern society, acting as the backbone for smart infrastructure, healthcare monitoring, and environmental sensing. However, the massive deployment of battery-operated devices poses a significant challenge regarding energy consumption and electronic waste. This dissertation contributes directly to the efficiency and sustainability of these systems.

The specific Sustainable Development Goals addressed by this work are:

SDG 7 Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable and modern energy for all. By optimizing the power consumption of data converters, we extend the battery life of devices and reduce the overall energy footprint of IoT networks.

SDG 9 Industry, Innovation and Infrastructure: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation. This work advances the state-of-the-art in microelectronics by proposing novel asynchronous architectures that enable smarter and more efficient industrial sensors.

SDG	Target	Contribution	Performance Indicators and Metrics
7	7.3	By double the global rate of improvement in energy efficiency, this work reduces the power waste in standby modes of electronic sensors.	Reduction in Energy per Conversion (fJ/conv) compared to synchronous architectures.
9	9.5	Enhance scientific research, upgrade the technological capabilities of industrial sectors, encouraging innovation.	Successful validation of the proposed calibration algorithm and circuit topology.

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List of Acronyms

IoT	Internet of Things
ADC	Analog-to-Digital Converter
DAC	Digital-to-Analog Converter
CMOS	Complementary Metal-Oxide-Semiconductor
LSB	Least Significant Bit
MSB	Most Significant Bit
SNDR	Signal-to-Noise-and-Distortion Ratio
SNR	Signal-to-Noise Ratio
SQNR	Signal-to-Quantization-Noise Ratio
ENOB	Effective Number of Bits
DNL	Differential Non-Linearity
INL	Integral Non-Linearity
BER	Bit Error Rate
FoM	Figure of Merit
SAR	Successive Approximation Register
LCS	Level-Crossing Sampling
GS/s	Giga-Samples per Second
MS/s	Mega-Samples per Second
SHA	Sample-and-Hold Amplifier
V_{th}	Threshold Voltage
V_{OS}	Input Offset Voltage
V_{DD}	Supply Voltage
INC	Increment event signal (upper level crossing)
DEC	Decrement event signal (lower level crossing)
DOUT	Digital output code
DOUT+	Upper digital threshold code
DOUT-	Lower digital threshold code
LC	Level Crossing
LFP	Local Field Potential
AP	Action Potential
RMSE	Root-Mean-Square Error

Chapter 1

Introduction

This chapter introduces the topic of the dissertation. It presents the context, the research question and the motivation for the work, states the objectives and the scope, and ends with the structure of the document.

Smart devices are now everywhere, a development usually grouped under the name Internet of Things (IoT). The processing and storage of information are digital and have benefited from the scaling described by Moore's law, but the physical world is analog: temperature, sound and biological potentials are continuous in both time and amplitude. The Analog-to-Digital Converter (ADC) is the interface between the two domains, and every digital system that measures the real world depends on one [66].

Energy efficiency has moved from a secondary performance metric to a primary design constraint. Many sensors run on small batteries or on harvested energy, so every joule dissipated shortens the autonomy of the system. The data conversion block is frequently the largest consumer in these interfaces, particularly when the digital transmission is duty-cycled and the radio spends most of its time off.

Many environmental and physiological signals are also sparse. Electrocardiograms, voice activity and environmental monitoring data arrive in bursts: short periods of high information content followed by long intervals in which nothing changes.

Traditional data conversion, based on uniform sampling at the rate dictated by the Nyquist-Shannon theorem [119], treats these silent periods exactly like the active ones. The system dissipates power to generate redundant samples that carry no new information. Exploiting this sparsity is the most direct route to ultra-low-power conversion interfaces.



Figure 1.1: Time-sparse signal representation and sampling strategies. (a) Time-sparse signal example showing burst activity with periods of inactivity. (b) Continuous-time event-driven sampling and conversion, which responds to signal changes. (c) Conventional uniform sampling operating at fixed Nyquist rate regardless of signal activity. (d) Discrete-time sampling with event-driven conversion, combining periodic and event-based approaches. This illustration demonstrates the efficiency of adaptive sampling methods for sparse signals compared to uniform sampling at the Nyquist rate [83].

Figure 1.1 shows this behaviour: the information in the burst waveform is concentrated in short intervals separated by long idle periods. Subfigures (b) and (d) show that continuous-time and discrete-time event-driven strategies generate no samples during inactivity, whereas conventional uniform sampling keeps running at a fixed Nyquist rate regardless of the signal content.

1.1 Motivation and Problem Statement

The energy wasted in large IoT deployments, together with the poor match between synchronous architectures and sparse data, motivates converters that activate only when the input actually produces an event.

1.1.1 Inefficiency of Synchronous Sampling

Traditional Analog-to-Digital Converters sample at a fixed rate set by a global clock (f_s). The Nyquist-Shannon sampling theorem [119] ties this rate to the maximum bandwidth of the signal, $f_s \geq 2B$. In many real applications, however, the signal is active for only a fraction of the time, so its true information content is far below what the bandwidth alone suggests.

The cost of this mismatch follows from the expression for dynamic power consumption in CMOS circuits [9]:

$$P_{dynamic} = \alpha \cdot f \cdot C_L \cdot V_{DD}^2 \quad (1.1)$$

where:

- $P_{dynamic}$ is the dynamic power dissipated.
- α is the activity factor (or switching factor).
- f is the clock frequency.
- C_L is the load capacitance being switched (e.g., in the clock tree).

- V_{DD} is the supply voltage.

In a uniformly clocked ADC, f is fixed by the Nyquist criterion and the clock network toggles every cycle, so α stays high even when the input is almost static. The converter keeps charging and discharging the clock and comparator capacitances at every period to produce samples that carry no new information. If a sensor signal is active only 10% of the time and the ADC samples continuously, 90% of the switching events are spent on intervals where the input is essentially constant.

1.2 Objectives and Scope of the Dissertation

Given the limitations of synchronous data conversion for time-sparse sensor signals, the primary objective of this work is to design and implement an ADC that exploits signal inactivity to reduce power consumption.

This overall goal breaks down into four specific objectives:

1. Asynchronous Architecture Design: To develop the circuit-level design of an asynchronous Flash ADC.
2. Offline Trimming Implementation: To implement an offline trimming mechanism capable of compensating the deterministic trip-point deviations of the comparison stage across process corners.
3. Performance Benchmarking: To quantify the energy efficiency of the proposed design using well known figures of merit. The results must be compared against the current state of the art synchronous ADCs and relevant asynchronous solutions.

1.3 Document Structure

This dissertation is organised into seven chapters. Chapter 1 introduces the topic, presents the motivation, problem statement, defines the objectives and scope of the work. Chapter 2 reviews the relevant background on analog to digital conversion, sampling paradigms, synchronous and asynchronous architectures, clockless event generation, calibration and trimming techniques, providing the basis for the proposed approach. Chapter 3 describes the design of the asynchronous Flash ADC, including the design specifications, the successive architectural choices and the final adopted solution. Chapter 4 presents the body bias trimming strategy adopted to correct the trip-point deviations of the comparison stage across process corners, building directly on the design of Chapter 3. Chapter 5 presents the synchronous reference Flash ADC used for comparison. Chapter 6 presents the simulation results and the performance evaluation of both converters, including the power measurement methodology used in the comparison. Finally, Chapter 7 summarises the main conclusions and outlines directions for future work.

Chapter 2

Literature Review

This chapter presents the state-of-the-art in Analog-to-Digital Converters, reviewing theoretical foundations and analysing survey data to justify the proposed architecture.

Subsequent sections focus on asynchronous and level-crossing sampling strategies, including prior work on event-driven converters and their advantages for time-sparse signals. The chapter closes with a review of calibration and trimming techniques for comparator non-idealities, which is the reference framework for the bulk-driven body bias trimming mechanism proposed later in this work.

2.1 Fundamentals of Analog-to-Digital Conversion

This section provides the theoretical background required to understand the operation and performance characterisation of data converters. It covers the fundamental steps of the conversion process such as sampling, quantisation, and coding.

2.1.1 The Data Conversion Process

2.1.1.1 Ideal Data Conversion

Analog-to-digital conversion bridges the continuous physical world and the discrete digital domain. It involves two distinct operations: discretisation in time (sampling) and discretisation in amplitude (quantisation). Ideally the process is instantaneous and lossless within the signal bandwidth of interest [66].

2.1.1.2 The Sampling Operation

Sampling converts a continuous-time signal $x(t)$ into a discrete-time sequence $x[n]$ [66].

According to the Nyquist-Shannon sampling theorem [119], a band-limited signal with maximum frequency B can be perfectly reconstructed if the sampling frequency f_s satisfies:

$$f_s \geq 2B \tag{2.1}$$

Violating this condition results in aliasing: high-frequency spectral components fold into the base-band and become indistinguishable from the original signal.

2.1.1.3 The Quantisation Operation

While sampling discretises the time axis, quantisation maps the continuous amplitude of each sample to one of a finite number of levels [66]. An N -bit ADC divides the input range (V_{ref}) into 2^N discrete levels. The step size between adjacent levels is the Least Significant Bit (LSB):

$$V_{LSB} = \frac{V_{ref}}{2^N} \quad (2.2)$$

Unlike sampling, quantisation is not reversible and introduces a deterministic error. This error, the quantisation noise, is usually modelled as additive white noise with a uniform probability distribution. For an ideal quantiser, the Signal-to-Quantisation-Noise Ratio (SQNR) is:

$$SQNR_{dB} \approx 6.02N + 1.76 \quad (2.3)$$

2.1.1.4 Coding

The final stage encodes the quantised level into a binary format. The most common schemes are straight binary and Gray code. The choice depends on the system requirements for data processing and transmission and does not affect the analog performance [9].

2.1.2 Classical Performance Metrics

A standard set of metrics, divided into dynamic and static parameters, allows data converters to be evaluated and compared objectively.

2.1.2.1 Resolution and Sampling Rate

Resolution defines the theoretical dynamic range, while the sampling rate determines the maximum signal bandwidth. Both are nominal values; the actual performance is limited by noise and non-linearities [66].

2.1.2.2 Signal-to-Noise-Distortion Ratio (SNDR)

The signal-to-noise-and-distortion ratio (SNDR) is the primary dynamic metric. It is the ratio of the signal power to the total power of all noise and harmonic distortion components. From SNDR, the Effective Number of Bits, ENOB, is derived [66]:

$$ENOB = \frac{SNDR_{dB} - 1.76}{6.02} \quad (2.4)$$

This value represents the true resolution of the converter at a specific input frequency.

2.1.2.3 Differential and Integral Non-Linearity (DNL and INL)

Static linearity is characterised by measuring the deviation of code transition levels from their ideal positions [66]. The DNL measures the deviation of a single step width from the ideal 1 LSB. A DNL less than -1 LSB implies a missing code in the transfer function.

The INL is the cumulative sum of DNL errors, representing the deviation of the transfer curve from a straight line. Specific INL patterns like saw-tooth shapes can reveal systematic errors in the architecture, such as gain mismatch or non-linear biasing [105].

2.1.2.4 Offset and Gain Error

These are linear errors: offset is a constant shift of the transfer characteristic, while gain error is a deviation in its slope [66]. Unlike non-linearity, both preserve the signal shape and can usually be calibrated out with little effort. Figure 2.1 shows their effect on the transfer characteristic. A constant DC shift moves the whole curve up or down (offset error), while a slope deviation alters the full-scale span (gain error). Both can be compensated by simple digital calibration.



Figure 2.1: Adapted from [96] ADC transfer function showing offset and gain errors. Offset error represents a constant DC shift applied uniformly across all codes, while gain error modifies the slope of the transfer function. Both are linear errors that can be calibrated using digital post-processing techniques [66].

2.1.2.5 Figures of Merit

Figures of merit (FoMs) provide normalised metrics to compare ADC performance across different architectures and technologies. The Walden FoM quantifies energy efficiency per conversion step:

$$\text{FoM}_W = \frac{\text{Power}}{2^{\text{ENOB}} \cdot f_s} \quad (2.5)$$

where lower values indicate better performance.

The Schreier FoM extends this by incorporating bandwidth effectiveness for oversampled converters:

$$\text{FoM}_S = \text{SNDR} + 10 \log_{10} \left(\frac{\text{BW}}{\text{Power}} \right) \quad (2.6)$$

with higher values representing superior performance.

2.1.3 Metrics Adaptation for Asynchronous Systems

The figures of merit defined above do not transfer directly to event-driven converters. In a level-crossing architecture, whether continuous- or discrete-time, the sampling rate is not a design constant: it depends on the activity and slope of the input signal. A formulation built around a fixed Nyquist frequency f_s therefore misrepresents the efficiency of these topologies.

The Walden FoM assumes a time-invariant sampling frequency. Level-crossing quantisers, as Weltin-Wu and Tsiividis demonstrated, generate a sample only when the input crosses a quantisation threshold [142]. When the input is sparse, few redundant samples are produced, and the saving carries through the whole downstream digital processing chain. The absence of a uniform sampling clock also changes the noise spectrum: the output is free of aliasing and of the usual quantisation noise floor, and under some signal conditions the SNDR can exceed the theoretical limit of a conventional converter with the same nominal number of bits.

A fair evaluation therefore replaces the fixed f_s in the Walden equation with the average rate of actual conversion events per unit time, $f_{s,avg}$. The discrete-time level-crossing ADC of Mou *et al.* shows why this matters: its power consumption varies between 95 nW for highly time-sparse stimuli, such as neural signals, and 250 nW for generic waveforms, so no single static number describes its efficiency [99].

With this adaptation, energy efficiency is reported in femtojoules per actual conversion step, and as a range rather than a single value. The prototype of Mou *et al.* spans 5.3 to 14.2 fJ per conversion step depending on the transition pattern of the input.

The variation has a physical origin. The amplitude change between successive samples, ΔV_{in} , determines how much work the converter must do. Slow events, with steps between one and two LSB, require only a linear ± 1 LSB update of the feedback loop and few comparisons. Abrupt variations beyond two LSB trigger a binary search similar to a SAR conversion, with more comparisons and more energy per event.

Throughout this work, the figures of merit of asynchronous converters are therefore indexed to $f_{s,avg}$, so that signal sparsity appears as a genuine efficiency gain instead of being hidden by uniform-sampling assumptions.

2.2 Sampling Paradigms: Synchronous and Asynchronous

Data conversion systems have traditionally relied on periodic sampling. An alternative is to let the dynamics of the input signal dictate when conversion happens. This choice separates synchronous from asynchronous operation.

2.2.1 Uniform Sampling and Nyquist Rate Systems

Uniform sampling follows the Nyquist-Shannon theorem: a constant sampling frequency of at least twice the highest frequency component of the signal. A clock distribution network times the sample-and-hold circuits and the quantisation stages that follow. The equally spaced temporal grid gives predictable throughput and simplifies digital signal processing. The drawback is that the converter spends dynamic power continuously, whether or not the input contains new information; Section 2.7 quantifies this overhead.

2.2.2 Non Uniform Sampling and Event Driven Systems

Non-uniform sampling removes the periodic clock altogether: conversion is governed by amplitude variations rather than by fixed time intervals. Level-crossing ADCs are the clearest example. A sample is generated only when the input crosses one of a set of predefined thresholds, so the effective sampling rate follows the instantaneous slope of the signal. Weltin-Wu and Tsividis [142] showed that the sample rate then becomes proportional to input activity, and power consumption follows it: the active components stay quiescent during low activity and dissipate dynamic power only when a conversion event occurs. The resulting representation of the continuous-time signal is also free of aliasing [142]. Discrete-time level-crossing architectures, such as the implementation by Mou et al. [99], add an adaptive search that switches between linear and binary tracking according to whether the comparator delay classifies each crossing as slow or fast.

2.2.3 Signal Sparsity and Theoretical Energy Efficiency Benefits

Signal sparsity is intrinsic to many physical phenomena, particularly in biomedical sensing, where physiological waveforms alternate long quiet intervals with sporadic bursts of high-frequency information. A uniform-sampling system must run at a Nyquist rate high enough to capture the bursts, and then wastes that effort during the silences. Event-driven architectures adapt the conversion rate to the information rate of the source: when the input is static, conversion activity drops to zero. Power dissipation is thus coupled directly to signal activity, which is what makes the nanowatt-level consumption reported by Weltin-Wu and Tsividis [142] and Mou et al. [99] possible.

2.3 Synchronous Architectures

Synchronous ADCs are the necessary starting point for any discussion of asynchronous ones. In these converters a global clock dictates the sampling instants and synchronises the internal operations. The architectures are mature and widely used, but they face power-efficiency trade-offs at high frequency and when the data is sparse [66].

2.3.1 Synchronous Flash ADC

The Flash ADC is conceptually the simplest and fastest architecture, performing a complete conversion in a single clock cycle [9]. It utilises a resistive ladder to generate $2^N - 1$ reference voltages, which are compared simultaneously to the input signal by a large bank of comparators. This parallel comparison produces a thermometer code, that a digital logic block then converts into a standard binary output. Its primary limitation is the exponential increase in hardware complexity, area, and power consumption as resolution increases.

The thermometer term refers to the parallel output from the Flash ADC comparators, where '1's run up to the input voltage level like mercury in a thermometer, indicating signal strength before being converted to standard binary by an encoder.

2.3.1.1 The Power Bottleneck

The primary drawback of the synchronous Flash architecture is its exponential scaling. Survey data clearly illustrates that high speed translates into high power in these designs. For example, a 6-bit 500 MS/s CMOS Flash ADC reported in 1999 already consumed 400 mW [126].

More extreme cases, such as a 22 GS/s 5-bit design, can consume up to 3W [114].

The continuous clocking of a massive comparator bank creates a significant energy dissipation even when the input signal is static. This lack of adaptivity makes synchronous Flash architectures increasingly unsuitable for battery-powered or energy-harvesting applications.

2.3.2 SAR (Successive Approximation Register)

The SAR ADC operates using a binary search algorithm [105], for each sample the internal logic tests one bit at a time, starting from the most significant bit (MSB). A single comparator compares the input to the output of an internal digital to analog converter (DAC), if the input is higher, the bit is set to '1' otherwise, it is set to '0'. This process repeats for N cycles. Because it uses very few active components, it is the most energy efficient choice for low speeds.

The binary search operation of a SAR ADC is implemented by successively testing each bit through the interaction of the sample-and-hold amplifier (SHA), DAC, comparator, and control logic, as shown in Fig. 2.2. At every conversion step, the SAR logic updates the DAC output and the comparator decides whether the input is above or below the trial voltage, gradually converging from the MSB to the LSB towards the final digital code.

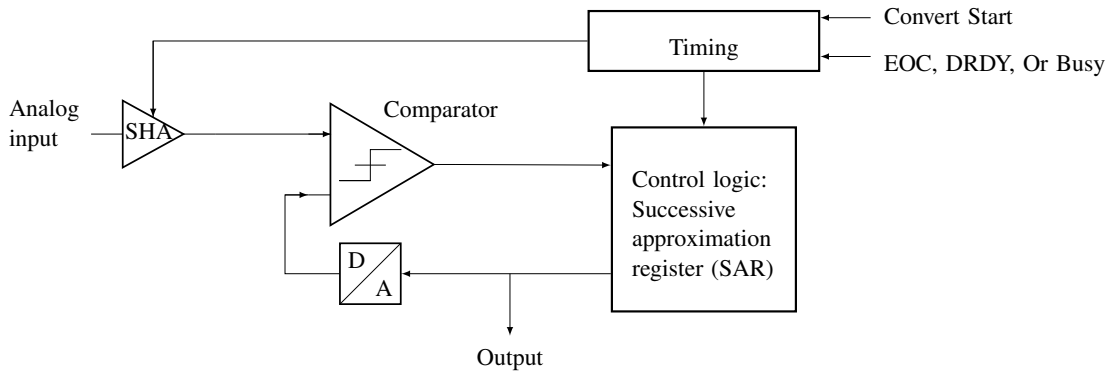


Figure 2.2: Successive Approximation Register (SAR) ADC architecture showing the sample-and-hold amplifier (SHA), comparator, digital-to-analog converter (DAC), and control logic. The SAR logic iteratively approximates the input signal by testing each bit from MSB to LSB, achieving high energy efficiency at moderate speeds [105].

2.3.3 Pipeline

The Pipeline ADC breaks the conversion into several sequential stages [66]. Each stage resolves a few bits, quantises them, and passes the amplified remaining error (the residue) to the next stage. Because several samples are processed concurrently, the architecture reaches high throughput at high resolution; the price is the processing latency.

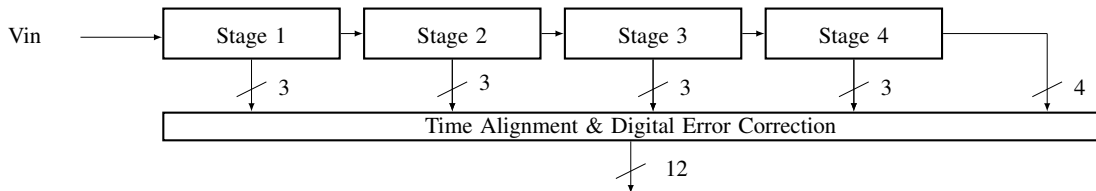


Figure 2.3: Pipeline ADC architecture showing multiple conversion stages operating in parallel on different input samples. Each stage resolves a subset of bits, generates a partial digital output, and amplifies the residue for the next stage. The pipelined structure enables high-throughput operation at the expense of increased latency and circuit complexity.

The pipelined converter processes multiple samples simultaneously by distributing the resolution across several cascaded stages, as shown in Fig. 2.3. Each stage resolves a subset of bits, generates a partial digital output, and amplifies the residue for the next stage, while a digital error-correction block aligns the timing of all stage outputs to produce the final high-throughput conversion result.

2.3.4 Sigma-Delta ($\Sigma\Delta$)

The $\Sigma\Delta$ architecture relies on oversampling, sampling much faster than the Nyquist rate [15], and noise-modulation. A modulator integrates the difference between the input signal and a feedback version of the quantised output. This process pushes the quantisation noise into higher frequencies, outside the signal band of interest, a digital filter then removes the high-frequency noise, resulting in high resolution and precision [113].

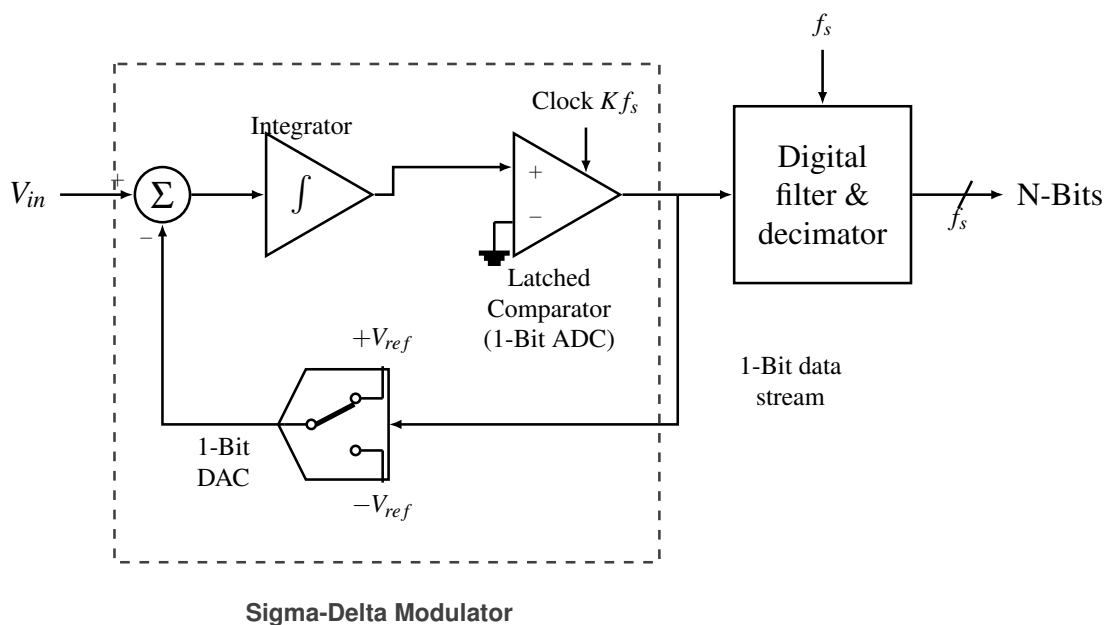


Figure 2.4: First-order sigma-delta ADC architecture illustrating the oversampling modulator with summer, integrator, 1-bit quantiser, feedback DAC, and digital decimation filter, which together implement noise shaping and high-resolution conversion [15, 113].

Oversampling and noise shaping are achieved by the feedback loop formed by summer, integrator, 1-bit quantiser, and digital decimation filter in the first-order sigma-delta architecture, as shown in Fig. 2.4. The modulator continuously integrates the difference between the analog input and the 1-bit DAC feedback, pushing most of the quantisation noise to high frequencies, which are later removed by the digital filter.

2.3.5 Dual-slope

This integrating architecture [111] measures the time a capacitor takes to charge and discharge. In the first phase the capacitor is charged by the input voltage for a fixed period; in the second it is

discharged by a known reference voltage. The time needed to return to zero is proportional to the average value of the input. The technique is accurate and immune to high-frequency noise, but much slower than the other architectures.

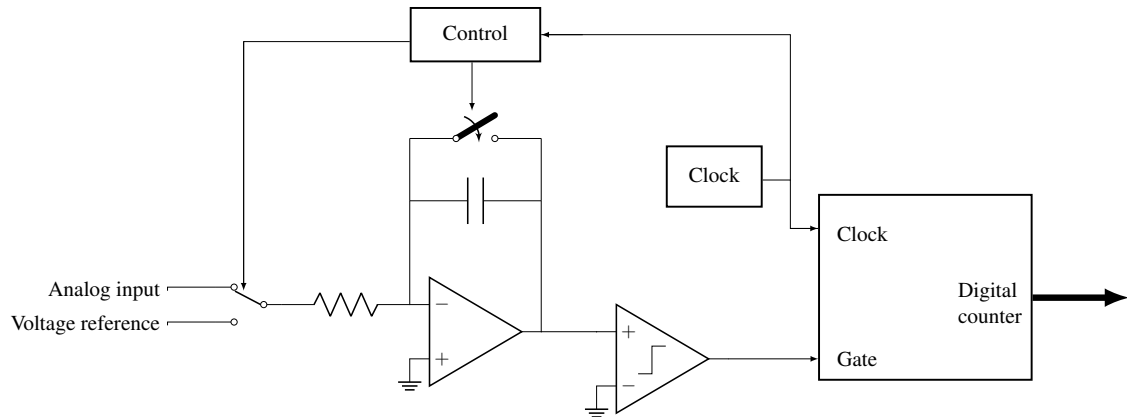


Figure 2.5: Dual-slope (integrating) ADC architecture showing the charging phase where an unknown input signal is integrated for a fixed time, and the discharge phase where a precise reference voltage drives the capacitor back to zero. The conversion time is proportional to the input voltage, providing excellent accuracy and noise rejection at the expense of low throughput.

The dual-slope architecture converts the input voltage into time by integrating it during a fixed interval and then discharging the capacitor with a known reference, as shown in Fig. 2.5. The digital counter measures the discharge interval controlled by the reference voltage, so the resulting count is proportional to the average value of the input signal and inherently rejects high-frequency noise.

2.4 Asynchronous Architectures

Asynchronous, or event-driven, architectures represent a fundamental paradigm shift in data conversion [83], unlike synchronous ADCs, which are bound by a global clock and the Nyquist-Shannon sampling theorem, asynchronous ADCs operate based on the signal's activity. This approach offers a more efficient alternative for processing sparse signals.

2.4.1 Level-Crossing Sampling (LCS)

The core principle behind many asynchronous ADCs is Level-Crossing Sampling (LCS) [83]. In traditional uniform sampling, the signal is captured at fixed time intervals (T_s), and the amplitude is quantised. In LCS, the process is inverted: the amplitude levels are fixed (quantisation thresholds), and the ADC records the exact time at which the input signal crosses these thresholds.

This method is particularly powerful for signals that remain constant or change slowly over long periods. Instead of generating redundant samples that capture no new information, the LCS ADC remains idle, only producing a digital event when the signal effectively changes by more than the defined threshold.



Figure 2.6: Digital logic core of a level-crossing asynchronous ADC showing the two DACs that generate upper and lower thresholds around the input signal V_{IN} , the pair of comparators that produce the INC and DEC event signals, and the digital logic that updates the output code $DOUT$ and the time-stamp Δt based on each level crossing. The feedback paths from the digital logic to the DACs keep the thresholds centered around the most recent output level, enabling event-driven tracking of the input signal instead of uniform-rate sampling [102].

The core of the proposed event-driven conversion scheme is illustrated in Fig. 2.6. In this architecture, two on-chip DACs generate a pair of voltage thresholds placed symmetrically around the analog input signal V_{IN} , while two comparators continuously monitor the crossings of V_{IN} with respect to these levels. The upper comparator asserts the INC signal whenever V_{IN} rises above the upper threshold, whereas the lower comparator asserts the DEC signal when V_{IN} falls below the lower threshold.

Upon each level-crossing event, the digital logic block updates the encoded amplitude word $DOUT$ and the control words driving the DACs so that the thresholds are recentered around the new signal level. In addition, the logic measures the elapsed time since the previous event and

outputs a time-stamp Δt , thereby representing the signal as a sequence of non-uniform events $(DOUT, \Delta t)$ instead of uniformly spaced samples. This mechanism concentrates conversion activity in intervals where the input varies rapidly and naturally suppresses conversions during flat regions, enabling an asynchronous ADC operation whose power consumption scales with the actual signal dynamics rather than with a fixed global sampling clock.

2.4.2 Flash ADC

The Asynchronous Flash ADC adapts the parallel structure of a standard Flash ADC but removes the sampling clock entirely. In this architecture, the comparators are not latched by a clock, they operate in continuous time, constantly monitoring the input voltage V_{in} against the reference ladder. The event generation happens when the input signal crosses a threshold, the corresponding comparator changes its output state immediately, this transition triggers an asynchronous logic to generate a digital output pulse.

2.5 Clockless Event Conversion

This section reviews current-mode techniques for generating conversion events without a clock: voltage-to-current conversion and current comparator topologies. A caveat applies to all of this material. These techniques are the conceptual basis of the first design phase in Chapter 3, the current-mirror Flash configuration, which the final architecture later abandoned because the voltage headroom demanded by current-mode structures did not fit within the 0.8 V supply. The review therefore documents the starting point of the architectural exploration and the criteria by which the current-mode candidates were judged and, in the end, rejected.

2.5.1 The Voltage to Current Conversion Concept in Event Detection

A central challenge in asynchronous Analog to Digital Converters is the generation of conversion events without a global clock. In a synchronous architecture, periodic sampling imposes the temporal discretisation from outside; an asynchronous system instead needs a mechanism that detects meaningful changes directly in the analog input. Voltage-to-current conversion is one efficient way to build such a mechanism.

The idea is to translate variations of the input voltage into proportional current variations that current-domain circuits can monitor continuously. Current-based processing often allows faster operation, which suits event-driven architectures where response time and energy efficiency both matter.

In this context, the analog input signal is applied to a transconductance stage that converts the input voltage into a current proportional to its amplitude. This generated current is then compared against a reference current or a set of threshold currents that represent predefined quantisation levels. Whenever the difference exceeds a decision threshold, an event is generated and propagated to the digital control logic.

This mechanism turns changes in signal amplitude into discrete temporal events. The converter stays inactive while the signal varies little and activates only on relevant transitions, which ties power consumption directly to input activity.

2.5.2 Current Comparator Operating Principles

Current comparators are the fundamental decision elements used in event driven architectures operating in the current domain. Their purpose is to determine whether an input current is greater or smaller than a reference current and to generate a corresponding digital output that signals the occurrence of an event.

Unlike conventional voltage comparators, which rely on charging and discharging internal nodes until a voltage threshold is reached, current comparators process information directly through current differences. This characteristic enables reduced voltage excursions, lower parasitic charging requirements, and potentially faster response times.

In a typical implementation, two input branches receive the signal current and a reference current. The comparator evaluates the difference between these currents and amplifies the resulting imbalance until one of the output states becomes dominant. Depending on the adopted topology, this amplification may occur through current mirrors, regenerative feedback structures, resistive loads, or current conveyor stages.

The performance of a current comparator is usually evaluated according to several parameters including propagation delay, resolution, sensitivity, power consumption, and input impedance. In asynchronous event generation applications, propagation delay becomes particularly relevant since it directly determines the temporal precision of the generated conversion events.

An ideal current comparator should therefore combine low input impedance, fast regeneration, low static power consumption, and robust operation under process and temperature variations.

2.5.3 Topologies for Event Detection in the Literature

Several event detection topologies have been proposed in the literature to enable continuous monitoring of analog signals and asynchronous generation of digital events, differing mainly in the method used to compare currents and regenerate the output transition. The six representative structures detailed below span the trade-off space between speed, power consumption, implementation complexity, and robustness. For asynchronous Flash Analog to Digital Converters, the preferred solution should preserve continuous operation while minimising propagation delay and avoiding unnecessary static power dissipation.

2.5.3.1 Current-Mirror-Based Current Comparator

The current-mirror-based comparator represents the simplest class of current-mode comparators. In this architecture, the input current and a reference current are applied to two branches of a mirror network that generates a difference current, which is then converted into a voltage across a high-resistance node. This voltage variation is sensed by a subsequent amplifier or inverter,

producing a logic output that indicates whether the input is larger or smaller than the reference. The small-signal behaviour is governed by the output resistance of the mirror devices: a high intrinsic resistance allows small current differences to produce significant voltage variations, which improves sensitivity.

The main advantage of this structure is its simplicity, very small silicon area and low design effort, since it relies on basic current mirrors and standard logic inverters. It can also offer good static power characteristics if the bias currents are kept small, and the absence of complex feedback paths simplifies stability considerations. However, the high output resistance that improves sensitivity also leads to poor frequency response when driving capacitive loads, so the addition of inverters or buffers to obtain rail-to-rail output significantly degrades speed. Their propagation delay can vary strongly with process and temperature, making them usually less suitable for very high-speed asynchronous Flash ADCs unless carefully optimised.

2.5.3.2 Traff Current Comparator

The Traff comparator introduces a high-speed continuous-time current comparison scheme based on a source-follower input stage and a CMOS inverter with positive feedback. The input current is first converted into a voltage at a low-impedance node through a pair of source followers, this node then drives an inverting stage that regenerates the decision. The positive feedback loop keeps the input node from slewing rail-to-rail, so its voltage excursion is limited around a quiescent point and the effective input impedance remains low, which shortens the time required to develop the voltage difference that triggers the output transition. When the differential input current exceeds the resolution threshold, the latch-like action of the feedback rapidly pulls the output to one of the rails, providing a logic-compatible decision.

Among its main advantages, this architecture achieves a high speed, with reduced propagation delay for a given input current step and relatively low power consumption for high-speed operation. Furthermore, the absence of large high-resistance nodes improves the frequency response and makes the design attractive for high-bandwidth current-mode ADCs. On the other hand, the Traff comparator exhibits a deadband region for very small input currents in which both input transistors operate near cut-off, temporarily increasing input resistance and degrading response time making it not suitable for this application.

2.5.3.3 Lu Chen Current Comparator

The continuous-time comparator proposed by Lu Chen employs a CMOS complementary amplifier with resistive feedback, followed by two resistive-load amplifiers and CMOS inverters. The core stage uses a pair of complementary transistors operating in saturation, with a MOS transistor in the linear region acting as a feedback resistor between input and output nodes. Small-signal analysis of this structure shows that, for typical device sizing, both input and output resistances of the complementary amplifier are approximately inversely proportional to the sum of the transconductances of the input devices, which strongly reduces the voltage swing at the internal nodes

during a comparison. The following resistive-load amplifiers provide additional gain with relatively low static current, while the final inverters restore rail-to-rail logic levels.

This design achieves a favourable speed-to-power ratio because the reduced internal voltage swings translate into shorter response time for a given input current, while the resistive-load stages limit power consumption, particularly when the input current increases. Compared with other high-speed comparators, simulations reported by the authors show that the average delay for small currents is comparable to more complex solutions, but with lower or similar power, and without requiring external bias currents or voltages, improving process robustness. As a drawback, the achievable resolution remains dependent on the sizing of the feedback device and the effective transconductance of the input pair, so aggressive scaling of bias currents to save power can degrade sensitivity.

2.5.3.4 Hysteresis Current Comparator

The hysteresis current comparator studied by Ranjana Sridhar introduces a current gain stage with non-linear feedback and a CMOS inverter output. The gain stage consists of cascaded resistive amplifiers and an inverter enclosed by a pair of transistors that implement non-linear feedback, whose gates are driven by the output node while their sources sense the difference-node voltage. For small $|I_{\text{diff}}|$, both feedback transistors are off and the node behaves as a capacitive load around a quiescent operating point, whereas for larger positive or negative differences one of the feedback devices turns on, enhancing the loop gain around that point and accelerating the transition.

This non-linear feedback mechanism creates an effective hysteresis, once the output has switched, a finite current difference of opposite polarity is required to bring the internal node back to the decision boundary, which improves noise immunity and metastability behaviour. The use of multiple gain stages and feedback devices increases circuit complexity and area compared with simpler comparators, and the hysteretic behaviour must be carefully dimensioned so that the added memory effect does not introduce excessive offset or distort the transfer characteristic in applications where strictly monotonic behaviour is required.

2.5.3.5 Min and Kim Current Comparator

The Min and Kim comparator employs a resistive feedback network around a current-source inverting amplifier to reduce input and output impedances and improve dynamic response. The structure consists of several current-source amplifiers connected in cascade, with the first stage incorporating a resistor that feeds back a portion of the output voltage to the input node. In steady state, this feedback lowers the effective impedance seen at the input and output, thereby limiting the voltage swing required to represent a given current difference and shortening the time needed to reach a decision threshold at the amplifier output. A final CMOS inverter converts the amplified voltage into a digital logic level.

Simulation results reported in the literature indicate that this architecture achieves low propagation delay and good process immunity, with the resistive feedback helping to stabilise the

operating point and reduce sensitivity to parameter variations. Additionally, the reduced node swings make the comparator suitable for relatively high-speed applications as long as the bias currents are chosen appropriately. A key limitation, however, is that both speed and resolution are strongly dependent on the bias current of the current-source amplifiers: achieving very high resolution requires higher bias currents, which directly increases static power dissipation and degrades energy efficiency. Moreover, because the design relies on current-source loads rather than resistive loads, its power consumption does not decrease when the input current increases, unlike other comparators where dynamic behaviour can partially trade off with static consumption.

2.5.3.6 CC-II-Based Current Comparator

The CC-II-based comparator proposed by Veepsa Bhatia uses a current subtractor followed by a current conveyor of the second generation and a multi-inverter output stage. The current subtractor is implemented with a p-type current mirror that produces the difference between the input current and a reference current across a resistor, effectively converting the difference into a proportional voltage. This voltage is then applied to the low-impedance X-terminal of a CC-II, whose behaviour is characterised by a unity voltage gain from Y to X and current conveyance from X to Z, so that the current difference is sensed at X and reproduced at the high-impedance Z terminal. The inverter chain at the output converts the conveyor's analog voltage into full-swing logic levels suitable for digital processing.

A key feature of this architecture is the use of positive feedback from one of the inverters back to the Y terminal of the conveyor, so that small variations in the X-node voltage induced by the input current difference are reinforced through the CC-II action. This positive feedback shortens the response time and increases sensitivity to small current steps, allowing the comparator to achieve sub-5 μA resolution with nanosecond-level propagation delay in simulations. The number of devices is also moderate, and the CC-II realisation provides inherently low input impedance at X and high output impedance at Z. On the downside, the presence of the op-amp or amplifier block used to realise the CC-II, together with the feedback network, increases design complexity and power compared with very simple mirror-based comparators. Furthermore, the performance is sensitive to the bandwidth and stability of the conveyor implementation, which must be carefully compensated to prevent oscillations due to the positive feedback.

2.6 Calibration and Trimming Techniques

In high-speed Flash ADCs, the accuracy of the system is fundamentally limited by the precision of the comparators. While the architecture fundamentals were established in the previous sections, practical implementations must address the non-idealities of the fabrication process, specifically the Input Offset Voltage (V_{os}).

2.6.1 The Component Mismatch Problem

In deep sub-micron CMOS technologies, transistors that are drawn with identical dimensions on the layout will exhibit slight differences in their electrical parameters after fabrication. This phenomenon, known as mismatch, affects the threshold voltage (V_{th}) and the current gain factor (β) of the differential pair in a comparator [9].

According to Pelgrom's law [104], the standard deviation of the threshold voltage mismatch ($\sigma_{V_{th}}$) is inversely proportional to the square root of the transistor area ($W \cdot L$):

$$\sigma_{V_{th}} = \frac{A_{V_{th}}}{\sqrt{W \cdot L}} \quad (2.7)$$

where $A_{V_{th}}$ is a technology-dependent constant. This creates a trade-off: to keep the offset small without calibration, the transistors must be large, which increases parasitic capacitance and degrades the speed and power efficiency of the ADC. In practice, small transistors are used for speed and calibration corrects the resulting offset.

2.6.2 Calibration Classifications

Calibration techniques can be classified by timing and by domain [9]. In timing, foreground calibration interrupts normal operation to measure and correct errors, while background calibration runs continuously at the cost of considerable extra complexity. In domain, digital calibration corrects the output code mathematically, whereas analog calibration adjusts the circuit biasing or load conditions to cancel the offset at its source.

2.6.3 Trimming Techniques

2.6.3.1 Internal Loading

The internal resistive loading method involves placing a variable resistive network in parallel or in series with the output loads of the comparator stage. By digitally switching small resistors or MOS switches operating in the triode region in parallel with the load branch, the effective resistance R_L is modulated. If the ADC has an offset, the trimming network is adjusted to introduce an equal and opposite value to effectively zero the error. This technique is typically implemented using a binary-weighted bank of PMOS transistors as the variable resistance, where the digital control code is determined at startup and stored in a register.

2.6.3.2 Reference Ladder Trimming

The reference ladder trimming technique corrects comparator offset by using the main reference ladder as a calibration source [146]. In this architecture, a switching network selects specific voltages from the resistor ladder and applies them to the bulk terminals of the input transistors M1 and M2. The body effect then shifts the threshold voltage (V_{th}) of the differential pair and cancels the input-referred offset caused by process mismatch.

As demonstrated in the simulation results of the threshold voltage ($|V_t|$) versus the ladder tap voltage (LT), the threshold voltage of transistors M1 and M2 increases linearly from approximately 356.5 mV to 372 mV as the calibration voltage is adjusted. In contrast, the threshold voltage of transistors M3 and M4 remains constant at approximately 356.5 mV because their bulk terminals are not connected to the tuning network. This targeted adjustment of the bulk potential for the input differential pair provides a tuning range sufficient to compensate for random process variations and ensure the accuracy of the comparator trip points.



Figure 2.7: Bulk-driven resistive reference ladder trimming architecture used for comparator offset calibration. The preamplifier input pair (M1, M2) has its bulk terminals driven by a selectable tap of the main reference resistor ladder via the offset calibration circuit (OSCAL). A digital control word selects one of the ladder taps (R_{tap}) through a tree decoder and multiplexers, generating a calibration voltage V_{cal} that modulates the bulk node V_B . By shifting the bulk potential of M1 and M2 while keeping the load devices at a fixed substrate potential, the threshold voltage of the input pair is adjusted through the body effect, providing a tunable range to cancel process-induced input offset in high-speed Flash ADC comparators [146].



Figure 2.8: Threshold voltage V_{th} variation of input differential pair transistors M1, M2 and load transistors M3, M4 as a function of the resistive ladder tap voltage LT . The graph demonstrates the body effect as the bulk potential of M1 and M2 is modulated via the reference ladder, providing a calibration range to compensate for process-induced mismatch [146].

The effectiveness of the bulk-driven trimming mechanism is illustrated by the controlled change of the input pair threshold voltage with the ladder tap voltage, as shown in Fig. 2.8 and Fig. 2.7. As the calibration voltage LT increases, the threshold of $M1$ and $M2$ shifts linearly while $M3$ and $M4$ remain fixed, providing the tuning range required to compensate comparator offset caused by random process variations [146].

2.6.4 Advantages of Static Bulk-Driven Trimming for Asynchronous ADCs

Dynamic offset-cancellation techniques such as auto-zeroing rely on accurately timed clock phases ϕ_1 and ϕ_2 and on storage capacitors that must be charged and discharged periodically [9]. Static trimming techniques, whether implemented with resistive networks or with bulk-driven threshold adjustment as in [146], suit an event-driven architecture far better. Once the correction values are found during a foreground calibration phase, the trimming network stays static: no further switching or clocking is required, and the comparator bias conditions are held by DC conduction paths alone. The trimming circuitry therefore does not disturb the event-driven operation of the ADC, which should remain idle whenever the input is not crossing a quantisation threshold.

The bulk-driven variant has a second advantage: the correction acts on terminals outside the signal path. Driving the bulk of the comparison devices shifts their threshold voltage through the body effect without adding capacitance to the high-speed nodes, so the small-signal bandwidth and regeneration speed are preserved. Dynamic calibration schemes, by contrast, place sampling capacitors and switch parasitics at sensitive nodes, which slows the comparator response and limits the maximum sampling rate in high-speed Flash architectures [9]. These properties motivate the static, bulk-driven trimming strategy adopted in Chapter 4, where the body bias of the comparison stage corrects deterministic trip-point deviations across process corners.

2.7 Asynchronous vs. Synchronous

2.7.1 Advantage

The primary motivation for adopting asynchronous architectures is the direct relationship between signal activity and power consumption [83].

In a synchronous system, the clock tree and the comparators switch at every clock cycle, regardless of whether the input is changing, the power consumption is dominated by the fixed clock frequency [9]:

$$P_{sync} \approx f_{clk} \cdot C_L \cdot V_{DD}^2 \quad (2.8)$$

In an asynchronous ADC, the switching frequency is replaced by the event rate. If the signal is constant or slowly varying, the comparators and digital logic remain static, leading to the following proportionality:

$$P_{async} \propto Activity \times V_{DD}^2 \quad (2.9)$$

The energy consumed is therefore proportional to the information content of the signal. For sparse signals this is a large advantage: during periods of inactivity the converter draws close to no dynamic power.

2.7.2 Comparative Evaluation

State-of-the-art data show that asynchronous ADCs are among the most energy-efficient converters over a wide range of sampling rates, particularly at low and medium bandwidths [83].

The plots and tables below draw on a set of 140 published designs. All entries come from ISSCC and JSSC, where the reporting of data converter performance is consistent and the peer review is strict. Within these sources, only papers reporting both the supply voltage (V_{DD}) and the signal-to-noise-and-distortion ratio (SNDR) were retained, so that every point in the performance envelopes rests on documented electrical and dynamic measurements. To ensure transparency and reproducibility, the complete list of references used to generate these performance envelopes, covering Flash, SAR, Pipeline, and Sigma-Delta architectures, is provided in [1–8, 10–14, 16–21, 23–28, 30–33, 35–41, 43–61, 63, 64, 67, 68, 70–80, 82, 84–93, 95, 97, 98, 101, 103, 106–109, 115–118, 120–128, 130–141, 143–145, 147–166].

In each plot, the entry marked [1] is the Flash ADC with the best performance among all surveyed works. It marks the limit of the performance envelope reached by Flash converters in that figure and is the benchmark against which the SAR, Pipeline and Sigma-Delta entries are compared.

The evolution of the Walden Figure of Merit as a function of sampling rate for different ADC architectures is summarised in Fig. 2.9. This plot allows identifying the performance envelope

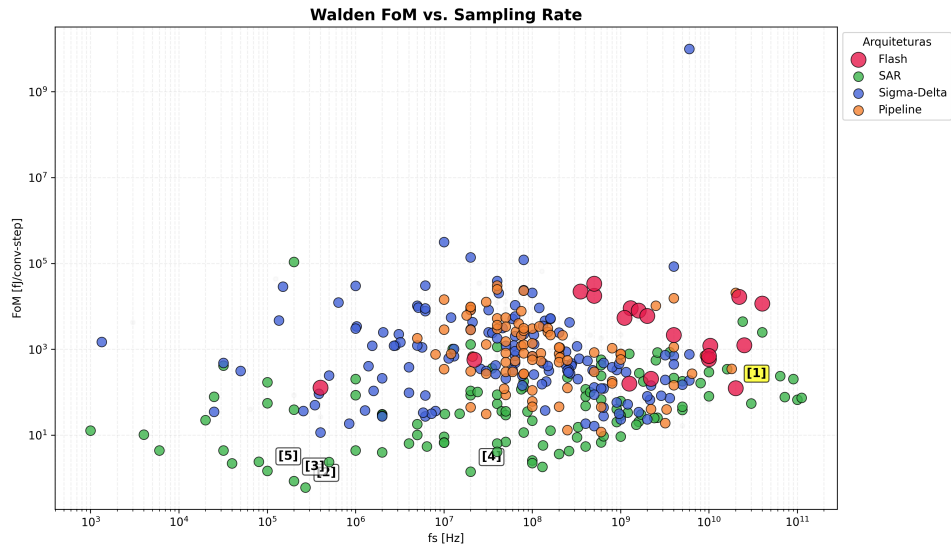


Figure 2.9: Walden FoM vs. Sampling Frequency (Flash reference [1] highlighted)

across a wide range of f_s and highlights how specific implementations approach the energy-efficiency limit at their target bandwidth.

Table 2.1: Walden FoM vs. Sampling Frequency (Flash reference [1] highlighted)

Ref.	Architecture	Publication	f_s [Hz]
[1]	Flash	Lukas [94]	20G
[2]	SAR	Jang [62]	270k
[3]	SAR	Ginsburg [42]	200k
[4]	SAR	Cano [29]	20M
[5]	SAR	Muller [100]	200

While Synchronous Flash ADCs are designed for peak performance at a specific frequency and suffer from a much higher power consumption caused by the continuous clocking, asynchronous designs scale their power consumption linearly with the input activity.

Table 2.2: Comparison of Power Consumption between Synchronous and Asynchronous Flash ADCs. Simulation results from ISSCC conference showing the power advantage of asynchronous operation for sparse signals.

Topology	Average power (P [W])
Synchronous	0.1016
Asynchronous	0.0314

Figure 2.10 shows how this architectural difference plays out over time. The plot tracks the

energy per conversion (P/f_s) over the last two decades. Both paradigms trend downwards with process scaling, but the asynchronous designs sit consistently at the lower bound of the energy envelope, which suggests that removing the global clock is one of the few remaining ways to push past the present efficiency limits.

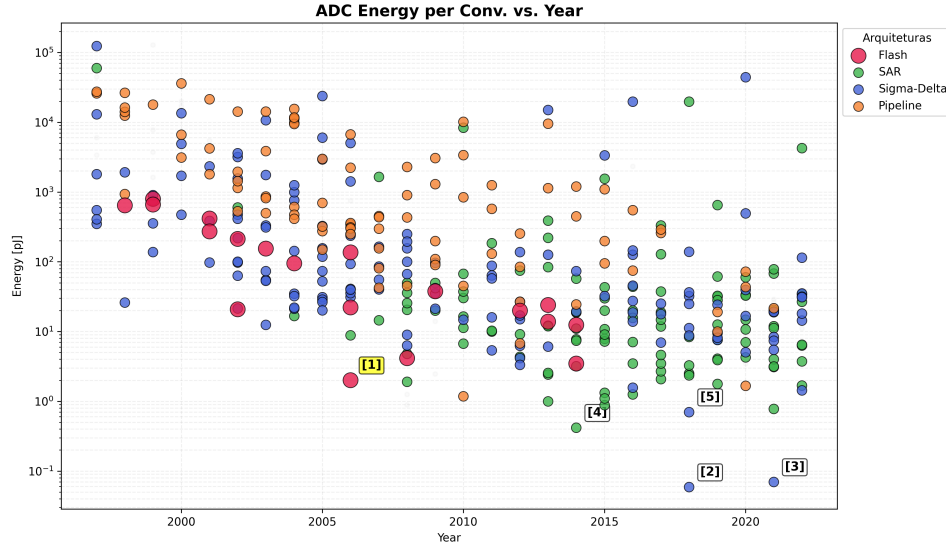


Figure 2.10: Energy per Conversion vs. Year (Flash reference [1] highlighted)

Each point in Fig. 2.10 corresponds to a published design, so the plot shows how the different ADC architectures fill the energy-versus-year plane and which solutions are the most efficient. The conclusion of this analysis is straightforward. Synchronous architectures waste power on

Table 2.3: Energy per Conversion vs. Year (Flash reference [1] highlighted)

Ref.	Architecture	Publication	Energy [pJ]
[1]	Flash	Chen [22]	2.02
[2]	Sigma-Delta	Jiang [65]	0.059
[3]	Sigma-Delta	Veldhoven [129]	0.070
[4]	SAR	Ginsburg [42]	0.420
[5]	Sigma-Delta	Kim [69]	0.703

time-sparse signals because the global clock keeps the switching activity, and hence the dynamic power, tied to the clock frequency even when the input carries nothing new. Event-driven schemes tie power to the rate of relevant signal transitions instead. The survey data adds the second half

of the argument: Flash ADCs define the speed frontier but pay for it with a poor Walden FoM, while the more efficient SAR and sigma-delta architectures are confined to lower bandwidths. An asynchronous Flash topology sits at the intersection of the two observations. By removing the sampling clock and letting the comparators operate in a level-crossing manner, it keeps the low latency of the Flash architecture and uses signal sparsity to close the efficiency gap with the best synchronous designs in the targeted operating region.

Event-driven operation is not exclusive to Flash. The discrete-time level-crossing converter of Mou et al. [99] consumes between 95 and 250 nW around a SAR-like structure, but it resolves each event through successive comparisons in a feedback loop, whereas a Flash bank resolves it in a single parallel step. For the low-latency, event-driven operation targeted here, that single-step conversion was the deciding argument in favour of Flash.

2.7.3 Chapter Summary and Design Directives

Asynchronous Flash ADCs combine a parallel comparison front-end with event-driven operation to exploit the time-sparse nature of many sensor signals. In these architectures, the conversion is triggered only when the signal crosses predefined decision levels, avoiding unnecessary activity during idle periods. Compared with conventional synchronous converters, this enables a significant reduction in dynamic power, since large capacitive nodes associated with sampling clocks and comparator arrays are only switched when relevant information is present in the input waveform.

From an architectural point of view, the proposed converter follows a Flash topology, where a bank of comparison elements observes different decision levels in parallel and produces a thermometer-like code that is subsequently encoded into binary. As in a conventional Flash ADC, this guarantees single-cycle conversion and very low latency, but here the global clock is removed and the decision activity is instead driven by input events. This choice transfers the power bottleneck from the sampling network to the design of the comparison elements themselves, since their intrinsic delay, input range and power consumption directly determine the achievable resolution, maximum event rate and overall energy efficiency.

For the comparison elements themselves, the current-mode techniques reviewed in Section 2.5 were the starting point of the design: processing the input as a current promises fast, clockless decisions, and the study of current comparator topologies shaped the first architectural phase in Chapter 3. That exploration will show, however, that the cumulative voltage headroom demanded by the current-mode structures did not fit the 0.8 V supply of the target technology. The final architecture operates in voltage mode instead, combining a complementary unity-gain voltage follower with a bank of ratioed CMOS slicers whose switching thresholds are set by transistor sizing. The literature in this chapter is therefore used in two ways: the current-mode material records where the first design phase came from and why it was rejected, and the trimming techniques of Section 2.6 underpin the body bias correction strategy of Chapter 4.

Chapter 3

Proposed Asynchronous Flash ADC Design

3.1 Design Specifications and Terminology

Before describing the architectural exploration, the specifications that constrained every design decision in this chapter are stated explicitly. The converter is designed in a 16 nm CMOS technology node, operating from a nominal supply voltage of $V_{DD} = 0.8$ V. The target application is the acquisition of time-sparse signals, as motivated in Chapter 1, which dictates the central architectural requirement: fully clockless, event-driven operation, with dynamic power dissipated only when the input crosses a decision level.

A point of terminology must also be fixed, since it is a recurring source of ambiguity in Flash converter descriptions. A Flash quantiser with a binary resolution of N bits requires $2^N - 1$ parallel comparison elements, whose collective outputs form a *thermometer code* of $2^N - 1$ levels; the number of thermometer outputs is therefore always distinct from, and larger than, the number of binary resolution bits. Throughout this chapter, the term *thermometer outputs* refers to the individual comparator or slicer outputs, while *resolution bits* refers exclusively to the binary word produced after encoding.

The converter implemented in this work has a resolution of 4 bits: a bank of $2^4 - 1 = 15$ slicers produces a fifteen-level thermometer code that the encoder maps to sixteen quantisation levels. This matches the resolution of the synchronous reference converter used for benchmarking (Chapter 5), which also uses fifteen comparators, so the two designs are compared at equal resolution. A presentational convention applies throughout the chapter: whenever the operating principle of the comparison stage is explained step by step, a reduced bank of *five* slicers (outputs out_4 to out_0) stands in for the full fifteen, purely to keep the figures and the code examples readable. The five-slicer bank is an explanatory device, not the implemented circuit; everything said about it extends directly to the fifteen-slicer bank of the actual design.

3.2 Architectural Evolution and Methodology

The design of the voltage comparator responsible for generating the digital output code of the analog-to-digital converter was not arrived at directly. It evolved through a sequence of candidate topologies, each evaluated against the constraints imposed by the target supply voltage, the required conversion accuracy, the effective input dynamic range, and the overall demands on circuit compactness. The following sections document this progression, presenting each topology that was considered and explaining the physical reasoning that either led to its adoption or prompted its replacement by a subsequent candidate.

3.3 Phase I: The Baseline Current-Mirror OTA Flash Configuration

The first topology investigated was the current-mirror flash ADC (CMF-ADC) introduced by Lee *et al.* [81], originally proposed as the sensing front end of a coarse-fine dual-loop digital low-dropout regulator. Although the application context of that work differs substantially from the present design, the CMF-ADC converts a voltage directly into a thermometer code, which is exactly what a multi-bit comparison stage needs, so it was a natural starting point for the architectural exploration. It is also the design phase that puts into practice the current-mode event-detection concepts reviewed in Section 2.5.

3.3.1 Operating Principle and Core Topology

The CMF-ADC topology is structured around two cascaded functional stages: a *transconductance cell*, which converts a differential input voltage into a proportional error current, and a *current-to-code converter*, which maps that current into a parallel thermometer-coded output without requiring a sampling clock. The complete circuit, annotated with the current-flow conventions used throughout this analysis, is shown in Figure 3.1.

3.3.1.1 Transconductance Cell

The transconductance cell is a differential pair formed by transistors M_1 and M_2 , both biased by a common tail current source of magnitude I_B . The non-inverting input of the pair receives the analog signal V_{in} , while the inverting input is driven by an adjustable reference voltage $V_{ref,c}$. An active current mirror, composed of diode-connected M_3 and mirroring transistor M_4 , constitutes the load of the differential pair and steers the drain current of M_1 toward the output. An additional NMOS mirror generates a third current branch I_3 as a replica of I_1 , which is then distributed to the conversion stage.

Equilibrium operation. See Figure 3.1(a) for the operation at equilibrium. When $V_{in} = V_{ref,c}$, M_1 and M_2 conduct equal quiescent currents of magnitude $I_B/2$. Because M_3 is diode-connected, its gate-to-source voltage adjusts to carry exactly $I_B/2$, and M_4 , sharing this gate voltage, mirrors the same current to the output branch. The mirrored current from M_4 and the drain current

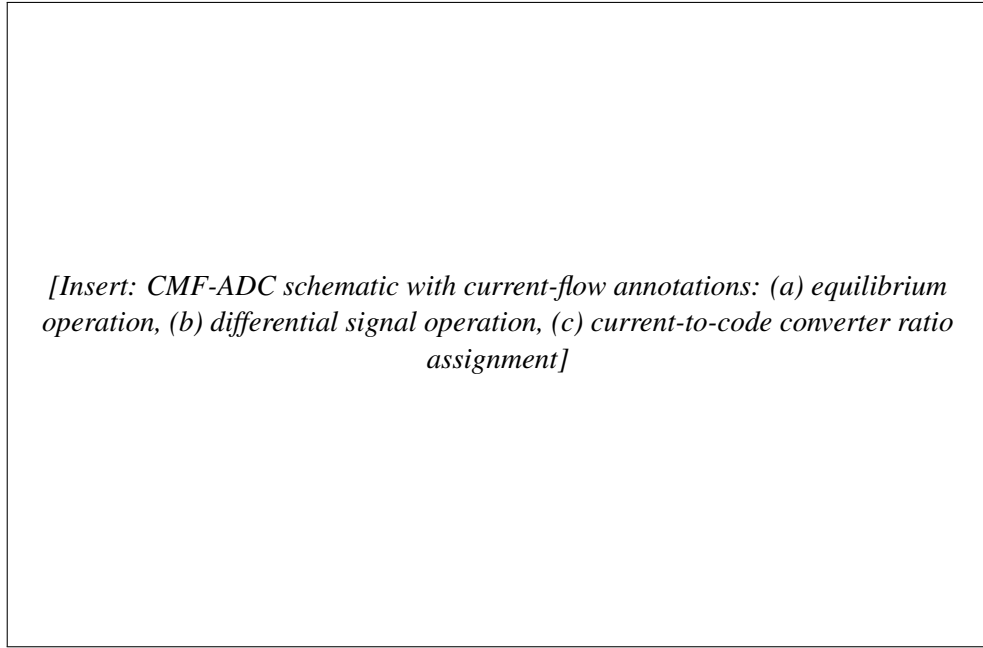


Figure 3.1: Operating principle of the current-mirror Flash ADC (CMF-ADC): (a) equilibrium operation, (b) signal operation, and (c) current-to-code conversion with asymmetric mirror ratios.

of M_2 are therefore equal in magnitude, so their contributions cancel at the output node of the transconductance cell and the net error current is zero: $I_{err} = 0$. Under this condition, as indicated in Figure 3.1(a), both branches carry identical currents in the same direction, resulting in no differential current available to drive the downstream converter.

Signal operation. See Figure 3.1(b) for the differential signal analysis. When a differential voltage $v_{id} = V_{in} - V_{ref,c}$ is applied at the input, the symmetry of the pair is broken. Assuming matched transconductances $g_m = g_{m1} = g_{m2}$, the two transistors steer the tail current unequally, producing branch currents:

$$I_1 = \frac{I_B}{2} - I_{err}, \quad I_2 = \frac{I_B}{2} + I_{err}, \quad (3.1)$$

where $I_{err} = g_m \cdot v_{id}/2$ is the error current. As annotated in Figure 3.1(b), when $V_{in} > V_{ref,c}$, M_1 carries the excess current (indicated by the thickened arrow on the M_1 branch), while the current in the M_2 branch is correspondingly reduced (indicated by the crossed symbol). The mirror formed by M_3 and M_4 copies I_1 toward the output, and the resulting imbalance between this mirrored current and I_2 constitutes the differential output signal $2I_{err}$ that drives the code converter. The effective transconductance of the cell is therefore:

$$G_m = g_{m1} = g_{m2} = g_m. \quad (3.2)$$

3.3.1.2 Current-to-Code Converter and Threshold Generation

Rather than producing a single differential output voltage, the current-to-code converter distributes I_2 and I_3 across N parallel comparison nodes, one per thermometer output. Each node is loaded simultaneously by a PMOS current mirror, which replicates I_2 with a multiplication factor α_k , and an NMOS current mirror, which replicates I_3 with a complementary factor β_k . The two factors are chosen such that α_k decreases monotonically with the node index k while β_k increases correspondingly, as illustrated schematically in Figure 3.1(c).

The binary output at node k is resolved by a simple current competition: if the PMOS sourcing current $\alpha_k I_2$ exceeds the NMOS sinking current $\beta_k I_3$, the node is pulled to a logic high, otherwise the NMOS pull-down dominates and the node settles to logic low. The quantisation threshold associated with node k is the specific input voltage at which the two competing currents are in equilibrium:

$$\alpha_k I_2 = \beta_k I_3 \Rightarrow v_{id}^{(k)} = \frac{I_B}{g_m} \cdot \frac{\alpha_k - \beta_k}{\alpha_k + \beta_k}. \quad (3.3)$$

Because the PMOS ratio α_k decreases monotonically with k , the thresholds $v_{id}^{(k)}$ also decrease monotonically, guaranteeing a valid thermometer code by construction. Nodes with a dominant PMOS mirror transition to high first as V_{in} rises above $V_{ref,c}$, followed progressively by nodes with higher NMOS ratios as the input continues to increase.

At the reference condition $V_{in} = V_{ref,c}$, the error current vanishes, $I_2 = I_3 = I_B/2$, and the state of each node reduces to comparing α_k against β_k directly. For an odd number of output bits with symmetric, evenly spaced ratio assignments, the central node satisfies $\alpha_k = \beta_k$ and is therefore at its decision boundary, producing the thermometer pattern $11 \cdots 1 ? 0 \cdots 00$ that is characteristic of a comparator array operating precisely at its reference level.

3.3.1.3 Adaptation to an Asynchronous Flash ADC

The circuit as described operates entirely in continuous time: no clock is needed to trigger a comparison, and the thermometer output updates immediately whenever V_{in} crosses one of the thresholds defined by Equation (3.3). This property is the primary motivation for adopting the CMF-ADC topology as the comparator front-end of an asynchronous Flash ADC.

In this context, each crossing event of V_{in} through a threshold level generates a spontaneous transition on the corresponding output bit without the intervention of any external timing reference. The ensemble of N output bits forms a thermometer code that tracks the input amplitude in real time: as V_{in} increases past each threshold, one additional bit transitions from low to high, and the reverse process occurs as V_{in} decreases. The asynchronous control logic described in Section 3.5.3 then detects these transitions and encodes them into the final binary output.

A further advantage in the ADC context is that monotonicity of the thermometer code is enforced at the topology level by the ordered mirror ratios, rather than relying on post-processing error correction. Bubble errors (isolated low codes within an otherwise high thermometer sequence) cannot arise as long as the PMOS ratios are strictly decreasing and the NMOS ratios are

strictly increasing across the output nodes, since any such ordering guarantees that the threshold voltages $v_{id}^{(k)}$ form a strictly monotone sequence as required by (3.3).

The main design parameters that control the ADC's resolution and input range are the mirror ratios $\{\alpha_k, \beta_k\}$, the tail current I_B , and the transconductance g_m of the input pair. These relationships, along with the specific ratio assignments used in the implementation explored in this work, are discussed in Section 3.3.2.

3.3.2 Reference Implementation and Mirror Ratio Assignment

The reference implementation described in [81] instantiates the operating principle of the previous section with five comparison nodes, i.e., five thermometer outputs (not five bits of binary resolution; see Section 3.1). The PMOS mirror ratios decrease monotonically across the five output nodes ($\times 14$, $\times 12$, $\times 10$, $\times 8$ and $\times 6$ for thermometer outputs $C_{LPT}[4]$ through $C_{LPT}[0]$), while the NMOS mirror ratios increase correspondingly from $\times 6$ to $\times 14$. The odd number of output nodes provides a well-defined, symmetric midpoint: the central node $C_{LPT}[2]$ carries equal PMOS and NMOS ratios ($\times 10$), so its decision threshold aligns exactly with the condition $V_{in} = V_{ref,c}$.

Applying Equation (3.3) to these ratio assignments confirms the expected behaviour at the reference condition: with $V_{in} = V_{ref,c}$, the two outermost node pairs resolve to high ($C_{LPT}[4]$, $C_{LPT}[3]$) and low ($C_{LPT}[1]$, $C_{LPT}[0]$) respectively, while the balanced central node sits at its decision boundary, yielding the thermometer code 11?00, where the question mark denotes the indeterminate state of a comparator operating precisely at its decision threshold.

The thresholds governing the transition of each output are fully determined by the mirror ratios, as established analytically in [81], and result in a monotonic, symmetric thermometer-coded response across the input voltage range. This architecture combines inherently fast conversion speed stemming from the absence of a conventional error amplifier, whose finite bandwidth would otherwise limit the response time of the control loop [81] with a straightforward, hardware-efficient structure and an explicitly quantifiable input-output characteristic. It was therefore adopted as the first candidate for the present comparator design, and its operating principle informed all subsequent topological choices.

3.4 Phase II: The Complementary Dual-Input Flash Configuration

3.4.1 Motivation for Wide Input Common-Mode Range Extension

One of the primary objectives during the development of the proposed comparator architecture was to extend its input common-mode range as close as possible to the supply rails. Achieving a rail-to-rail input capability is desirable because it allows the converter to correctly process analog signals whose common-mode voltage may assume any value within the available supply range, from ground to V_{DD} , without requiring additional level-shifting or biasing circuitry.

This characteristic is particularly important in mixed-signal sensor interfaces, where the output voltage of the sensing element is often determined by the physical phenomenon being measured

rather than by the requirements of the analog front-end. Restricting the admissible common-mode voltage consequently reduces the effective operating range of the converter and may require additional conditioning stages, increasing both power consumption and circuit complexity.

Furthermore, maximising the input common-mode range enables better utilisation of the available voltage swing. Since the signal occupies a larger fraction of the supply voltage, the effective dynamic range and signal-to-noise ratio can be improved without increasing the supply voltage itself. This consideration becomes especially relevant in modern low-voltage CMOS technologies, where the continuous reduction of V_{DD} inherently limits the available analog headroom.

A wide common-mode operating range also improves the robustness of the system against process, voltage and temperature (PVT) variations. Small shifts in the input common-mode voltage caused by fabrication mismatch or supply fluctuations are less likely to push the input transistors out of their intended operating region, so the transconductance is preserved and the comparator behaviour stays predictable.

For these reasons, a complementary dual-input architecture capable of providing rail-to-rail operation was investigated as a second design candidate.

3.4.2 Rail-to-Rail Architecture Evaluation and Voltage Headroom Limitations

Although the proposed CMF-ADC satisfies the functional requirements of the event-driven conversion scheme, its input common-mode range remains inherently limited by the voltage headroom required by the differential input pair and the tail current source. This limitation becomes particularly critical in modern low-voltage CMOS technologies, where the continuous reduction of the supply voltage severely restricts the available analog signal swing. As a consequence, input signals whose common-mode voltage approaches either supply rail may drive the transconductance stage outside its intended operating region, degrading both linearity and sensitivity.

The complementary dual-input topology evaluated to overcome this limitation employs two transconductance stages operating simultaneously: one based on an NMOS differential input pair and another based on a PMOS differential input pair. The rationale behind this arrangement is well established in rail-to-rail input amplifiers [110]. An NMOS differential pair operates correctly when the common-mode voltage is high enough above the negative rail to provide gate overdrive for the pair and saturation for its tail source, so it covers the upper part of the input range; a PMOS differential pair has the complementary behaviour and covers the region near the negative rail. With both stages operating simultaneously, at least one transconductor remains functional across almost the entire supply voltage range.

The implementation explored in this work did not simply combine the output currents of the two OTAs in parallel. Instead, one output branch of each OTA was connected to the PMOS current mirrors responsible for generating the individual bits of the thermometer code, preserving the current-comparison principle established in the original CMF-ADC architecture. The remaining output branch of each OTA was then cross-coupled to the complementary transconductor, such that the residual output current of one stage modulated the output node of the other, with the same arrangement applied symmetrically in the opposite direction.

The objective of this bidirectional cross-coupling was to exploit the complementary transconductance of both stages so that each OTA could reinforce the operation of the other near the boundaries of its valid input common-mode range. In principle, this mechanism would extend the effective dynamic operating region of the converter while preserving the event-driven current-comparison methodology.

Despite the conceptual advantages of this architecture, its practical implementation proved incompatible with the target supply voltage of $V_{DD} = 0.8$ V. A conventional five-transistor OTA already requires a minimum voltage headroom to accommodate the gate-to-source voltage of the differential pair, the saturation voltage of the tail current source, and the voltage drop across the active load. At a supply voltage of only 0.8 V, this headroom budget is already severely constrained.

The current-mirror OTA implementation did not fit this budget. In that topology, each load is itself a mirror that copies the branch current onward to the thermometer-code conversion stage, so every signal path stacks the gate-to-source voltage of the input pair, the saturation voltage of the tail source and the gate-to-source voltage of the mirror load; the cross-coupling mechanism then inserted one more transistor into this same path. Each stacked device adds its own overdrive or saturation voltage requirement, and the sum of these contributions exceeded the available supply. No operating point kept all active devices in saturation over the intended input range, so the architecture could not function correctly.

Part of the lost headroom could be recovered at the tail of the OTAs. The conventional tail current source was replaced with the low-voltage current mirror described in Razavi's book [110], which needs less voltage across the mirror branch because it removes one stacked gate-to-source drop from the path, at the price of a more complex bias network. The change helped, but not enough: the combined voltage requirements of the differential input pair, the low-voltage tail mirror, the mirror loads and the cross-coupled output stage still exceeded the available supply.

No feasible bias point could be identified that simultaneously maintained all transistors within their intended operating regions while preserving the desired dynamic-range extension mechanism. What this phase showed is that the current-mirror implementation of the rail-to-rail concept cost too much headroom at 0.8 V; the concept itself survived, along with the low-voltage tail mirror. The third design phase keeps the complementary dual-input idea and rebuilds it around a far smaller stack of transistors.

3.5 Phase III: The Final Adopted Architecture

3.5.1 Complementary Five-Transistor OTA Core and Optimised Transistor Sizing

The third design phase kept the rail-to-rail objective of Phase II and rebuilt its implementation around the classical five-transistor operational transconductance amplifier. The complementary input is preserved: two five-transistor OTAs operate in parallel on the same input, one built around an NMOS differential pair and the other around a PMOS differential pair, so the usable input range

is extended in the same way as before. The load structure is what changes. Each pair is loaded by a single active current mirror, in which one device operates in diode connection and sets the gate voltage of the mirrored branch; this is the conventional active load that defines the five-transistor OTA [110].

The headroom is recovered in this load. In the Phase II topology every load was itself a mirror that copied the branch current onward, so each signal path carried the extra stacked gate-to-source drops described in the previous section. With one current-mirror active load per pair, those extra devices and their accumulated overdrive requirements disappear, and the complementary structure fits within the $V_{DD} = 0.8$ V budget while it keeps the extended input range.

The tail bias carries over a result from the previous phase. Both OTAs draw their tail current through the low-voltage current mirror from Razavi's book [110], the same structure tested in Phase II, because even this reduced topology operates close to the headroom limit and a conventional tail mirror would spend voltage the design does not have.

One limitation remains: close to either supply rail neither input pair has enough overdrive to stay in saturation, so the effective input window of the complementary structure stops slightly short of the rails. The consequences for the test stimuli and for the idle power consumption are discussed in Section 3.6.2.

The initial implementation evaluated the complementary OTA under open loop conditions in order to maximise gain and improve sensitivity to small differential input voltages. Through transistor sizing optimisation, high voltage gain could be achieved and with it higher comparison sensitivity. Practical simulations, however, revealed a critical limitation of this approach: small unintended differential inputs originating from offset, thermal noise or marginal overdrive caused the output node to saturate rapidly at one of the supply rails before the intended comparison could complete.

Once saturation occurred, the recovery was limited by the bias current available to charge or discharge the high-impedance output node. This increased the settling time considerably and introduced variability between consecutive conversion cycles.

To overcome this limitation, the complementary OTA was reconfigured as a unity gain voltage follower by directly connecting the output node to the inverting input and establishing full negative feedback. Under this configuration, the differential input voltage remains close to zero in steady state and the amplifier operates continuously in its linear region.

As a consequence, the OTA acts as a low impedance replica of the input signal, reproducing V_{in} at its output while avoiding saturation effects. The resulting accuracy is determined primarily by the open loop gain and the input referred offset of the amplifier.

3.5.2 Ratioed Output Slicers Design

The quantisation stage of the converter is implemented through a bank of CMOS slicer inverters directly connected to the output of the voltage follower. Unlike the previously explored current mirror based architecture, where threshold positioning was determined by current scaling, the

present design defines each comparison level through the PMOS to NMOS sizing ratio of each inverter.

The switching threshold of a CMOS inverter is determined by the relative strengths of the pull up and pull down networks and can be approximated by

$$V_M \approx \frac{V_{DD}\sqrt{r}}{1 + \sqrt{r}} \quad (3.4)$$

where

$$r = \frac{(W/L)_p}{(W/L)_n}$$

represents the ratio between PMOS and NMOS transistor dimensions.

An increase in r shifts the switching threshold toward higher voltages, whereas lower values move the transition below the mid supply level. By assigning distinct transistor ratios to each inverter, a distributed set of decision thresholds is created along the input voltage axis.

Each slicer therefore acts as an independent comparator operating at a predefined threshold voltage. Collectively, the outputs generate a thermometer coded representation of the input signal.

The sizing strategy is illustrated here with the reduced five-slicer bank introduced in Section 3.1. In that explanatory configuration, the topmost thermometer output, out_4 , uses a transistor ratio of $r = 1.5$, which places its switching threshold above 0.4 V; the bottommost output, out_0 , adopts $r = 0.5$ and a correspondingly lower threshold; and the intermediate slicers are distributed between these limits to generate approximately symmetric threshold spacing around the reference level.

As the input voltage increases from ground toward V_{DD} , each inverter switches sequentially according to its assigned threshold. At $V_{in} = 0.4$ V, slicers with $r > 1$ remain in logic high state while slicers with $r < 1$ have already transitioned to logic low. The inverter designed with $r = 1$ operates exactly at its switching point and therefore enters a metastable region. The resulting thermometer code becomes 11?00, reproducing the same behaviour previously observed in the CMF ADC architecture.

The implemented converter applies exactly the same strategy to its full bank of fifteen slicers: each slicer receives a distinct PMOS-to-NMOS ratio, and the fifteen ratios place the fifteen decision thresholds across the input range required for 4-bit resolution.

3.5.3 Asynchronous Control Logic and Clockless Event Generation Network

A key characteristic of the proposed architecture is the complete removal of the global clock from the conversion process. Instead of forcing periodic operation, event generation occurs exclusively as a consequence of changes in the analog input signal.

The OTA configured as a voltage follower continuously tracks the input signal and distributes this low impedance representation to all slicers simultaneously. Each slicer independently evaluates whether the input has crossed its corresponding threshold and immediately updates its output state.

The collection of slicer outputs forms an asynchronous thermometer code whose transitions directly represent input activity. Whenever one or more thresholds are crossed, a digital event is naturally generated without requiring any external timing reference.

The digital backend that converts these transitions into a usable output code consists of three simple functions, all of them purely combinational or edge-triggered by the slicer outputs themselves, so that no timing reference is ever required. First, a per-slicer transition detector monitors each thermometer output and produces a short strobe whenever that output changes state in either direction. Second, the strobes of all slicers are merged by an OR network into a single conversion-event signal, which flags that the thermometer code has changed and that a new output word is available; this event signal also carries the timing information of the conversion, since in level-crossing operation the instant of the event is part of the encoded information. Recording those instants is the task of whatever system consumes the converter output, and its cost lies outside the converter core characterised in this work, in the same way as the encoder discussed below. Third, a thermometer-to-binary encoder maps the current state of the slicer bank into the output binary word; because the slicer thresholds are monotonically ordered by construction at the nominal process corner, this mapping reduces to counting the number of asserted thermometer outputs. Process variation can disturb that ordering, and restoring it is precisely the purpose of the body bias trimming of Chapter 4. In the simulation framework of this work, the transition detection and encoding functions were modelled behaviourally for the mixed-signal co-simulation. As discussed in Chapter 6, the encoder power is excluded from the reported figures of both the asynchronous and the synchronous converters, since an equivalent encoder is required in both.

This event driven operation establishes a direct relationship between signal activity and switching activity inside the converter. During intervals where the input remains constant, the slicers remain static and dynamic power consumption approaches zero. Conversely, periods of rapid input variation automatically increase event generation frequency and conversion activity.

Compared with previous current mirror based implementations, this architecture introduces a clear separation between input buffering and threshold generation. The OTA is responsible only for reproducing the input voltage, while threshold definition is achieved entirely through transistor sizing in the slicer stage.

This separation simplifies the design procedure, makes the circuit robust against saturation and removes the extra voltage headroom requirements. The result is a compact, scalable architecture suited to low-voltage asynchronous Flash conversion.

3.6 Analytical Modelling and Design Constraints

This section establishes the theoretical limits and design trade-offs of the asynchronous Flash ADC. In a synchronous architecture the performance limits follow from the clock frequency. In a level-crossing converter they depend instead on the dynamics of the input signal itself, on the circuit propagation delays and on the noise.

3.6.1 Maximum Input Wave Frequency and Slope Overload

In synchronous ADCs the maximum input frequency follows from the Nyquist-Shannon sampling theorem ($f_{in} < f_s/2$). An asynchronous Flash or level-crossing ADC has no uniform sampling grid; its limit is the maximum tracking speed of its internal circuitry, expressed as a slew rate (SR) constraint. When the input changes faster than the converter can track, the result is a distortion known as slope overload.

Consider a full-scale sinusoidal input voltage $V_{in}(t)$:

$$V_{in}(t) = \frac{V_{FS}}{2} \sin(2\pi f_{in} t) \quad (3.5)$$

where V_{FS} represents the full-scale voltage range of the converter and f_{in} is the input frequency. The time derivative of this signal yields its instantaneous slope, which achieves its maximum absolute value at the zero-crossing points:

$$SR_{max} = \left. \frac{dV_{in}(t)}{dt} \right|_{max} = \pi f_{in} V_{FS} \quad (3.6)$$

The asynchronous Flash ADC resolves the input signal by triggering a new state or quantisation conversion whenever the input crosses a discrete threshold voltage step Δ , corresponding to the nominal resolution of N bits:

$$\Delta = \frac{V_{FS}}{2^N} \quad (3.7)$$

Even in an ideal, noiseless scenario, each conversion takes a finite minimum time: the input follower must settle, the slicer associated with the crossed threshold must switch, and the event-generation logic must propagate the result. The sum of these contributions is the total conversion propagation delay τ_{conv} . Consequently, the maximum rate at which the ADC can track voltage transitions (the converter's intrinsic Slew Rate, SR_{ADC}) is bounded by:

$$SR_{ADC} = \frac{\Delta}{\tau_{conv}} = \frac{V_{FS}}{2^N \tau_{conv}} \quad (3.8)$$

To circumvent slope overload distortion, the maximum slope of the input signal must never exceed the tracking capability of the asynchronous circuitry ($SR_{max} \leq SR_{ADC}$):

$$\pi f_{in} V_{FS} \leq \frac{V_{FS}}{2^N \tau_{conv}} \quad (3.9)$$

Rearranging this inequality yields the absolute maximum input frequency $f_{in,max}$ that the converter can faithfully digitise without undergoing slope overload under ideal conditions:

$$f_{in,max} = \frac{1}{\pi 2^N \tau_{conv}} \quad (3.10)$$

Transient noise lowers this limit considerably. Superimposed thermal and substrate noise produces rapid local voltage fluctuations (dV_{noise}/dt) that inflate the effective slope of the signal and make the voltage bounce back and forth across a quantisation threshold. These bounces consume the processing window τ_{conv} and appear at the output as thermometer-code metastability, the so-called *bubble errors*. The practical maximum slew rate under noise is therefore well below the ideal value: slope overload appears at lower input frequencies and the measured bubble rate rises.

3.6.2 Resolution and Power Dissipation

The signal-dependent power profile is one of the main advantages of asynchronous data converters. In a standard synchronous Flash ADC the power consumption is roughly constant: the whole comparator bank ($2^N - 1$ devices) and the clock distribution network toggle at the sampling frequency f_s regardless of what the input is doing. In an asynchronous implementation, power dissipation follows the rate of change of the input.

When the input signal is entirely static ($f_{in} = 0$), no quantisation thresholds are crossed. Under this condition, the dynamic switching power ideally drops to zero, and the system only dissipates static power (P_{static}), which originates from the continuous bias currents in the analog references, pre-amplifiers, and comparators:

$$P_{total} = P_{static} + P_{dynamic} \quad (3.11)$$

In the implemented converter this zero-power condition is a theoretical limit rather than a measurable operating point. As discussed in Section 3.5.1, the complementary follower operates correctly only within an input window that stops slightly short of both supply rails, so the dynamic characterisation used a sinusoidal stimulus between 0.1 V and 0.7 V. Because the input never reaches the bottom rail, the input stages always conduct a residual bias current, and the measured consumption at idle settles at a non-zero floor set by P_{static} instead of falling to the ideal zero.

For a dynamic input, such as a full-scale sinusoid at frequency f_{in} , the signal traverses the full-scale range V_{FS} twice per period $T = 1/f_{in}$ (one upward and one downward excursion across the 2^N levels in each half). The average number of quantisation level crossings per second, which can be read as an effective average sampling frequency $f_{s,avg}$, is therefore proportional to both the frequency and the resolution:

$$f_{s,avg} = 2 \cdot 2^N \cdot f_{in} = 2^{N+1} f_{in} \quad (3.12)$$

Assuming that each individual asynchronous conversion cycle draws an average switching energy E_{conv} from the power supply V_{DD} , the dynamic power consumption $P_{dynamic}$ can be modelled

analytically as:

$$P_{dynamic} = E_{conv} \cdot f_{s,avg} = E_{conv} \cdot 2^{N+1} f_{in} \quad (3.13)$$

Substituting this back into the total power equation yields:

$$P_{total} = P_{static} + E_{conv} \cdot 2^{N+1} f_{in} \quad (3.14)$$

Equation (3.14) shows that $P_{dynamic}$ grows linearly with the signal frequency f_{in} but exponentially with the nominal resolution N .

In practical circuit implementations, the physical performance is constrained by noise, which splits the design analysis into nominal resolution (N) and Effective Number of Bits ($ENOB$), the latter derived directly from the Signal-to-Noise-and-Distortion Ratio ($SNDR$ or $SINAD$) as defined in Chapter 2.

Transient noise degrades the Signal-to-Noise Ratio (SNR) and the $SNDR$, and with them the $ENOB$. It also introduces spurious, non-monotonic threshold crossings (chatter), and each of these noise-driven crossings triggers a redundant conversion cycle that encodes no useful information. The bubble rate rises and $P_{dynamic}$ grows well beyond the ideal prediction. Power goes up while $ENOB$ goes down, so the Walden Figure of Merit (FoM_W) suffers twice over. Keeping the noise-induced bubble rate low is therefore a first-order design constraint for energy efficiency.

3.6.3 Chapter Summary

This chapter established the architecture of the proposed asynchronous Flash ADC and the theory that bounds its performance. The converter replaces clock-bound sampling with signal-dependent quantisation: with no global clock, it avoids the energy that synchronous architectures waste on sparse or bursty signals.

The structural development progressed through three architectural phases: a current-mirror Flash configuration adopted from the literature, a current-mirror implementation of the complementary dual-input extension rejected for exceeding the voltage headroom available at 0.8 V, and the final voltage-mode solution, in which a complementary pair of five-transistor OTAs configured as a unity-gain follower drives a bank of ratioed CMOS slicers whose switching thresholds are set purely by transistor sizing. The event-generation network described in Section 3.5.3 completes the conversion chain, detecting the transitions of the thermometer outputs and encoding them combinatorially into the binary output word, with no clock involved at any point in the signal path.

Finally, the analytical models bounded the design space. The maximum input frequency is set not by a Nyquist limit but by the slew rate constraint SR_{ADC} , in which the propagation delay τ_{conv} determines the onset of slope overload. The power analysis showed that dynamic consumption grows linearly with the input frequency f_{in} but exponentially with the nominal resolution N . The same models quantify the cost of transient noise: degraded SNR and $SNDR$, bubble errors, and redundant conversion cycles that raise the power while the effective resolution ($ENOB$) falls, to the

detriment of the FoM_W . These results are the baseline for the simulation setup and the performance evaluation of the following chapters.

Chapter 4

Body Bias Trimming Strategy

The slicer-based quantisation stage introduced in Chapter 3 defines each decision threshold exclusively through the PMOS-to-NMOS sizing ratio of a CMOS inverter, as expressed by equation (3.4). This makes the converter linearity directly dependent on the effective drive strengths of the two device types, and therefore directly exposed to fabrication process variability. This chapter describes the offline trimming strategy adopted to correct the resulting deterministic trip-point deviations across process corners. The technique builds on the static, bulk-driven calibration approaches reviewed in Section 2.6: it adjusts the threshold voltages of the slicer transistors through the body effect, in the same spirit as the bulk-driven reference-ladder trimming of Yao [146] discussed in Section 2.6.3.2, but applied here to the bulk terminals of the slicer bank and indexed by the identified process corner rather than by a per-comparator offset measurement. The scope of this correction is deterministic inter-corner variation only: random intra-die mismatch, governed by Pelgrom’s law [104], is not addressed by the corner-indexed table and is discussed as future work in Chapter 7.

4.1 Choice of Body Biasing Among the Reviewed Techniques

Among the trimming techniques surveyed in Section 2.6, several alternatives were considered before adopting body biasing. Internal resistive loading, in which a digitally switched resistor bank modulates the load of each comparator branch, is well suited to differential comparator stages but does not map naturally onto the slicer inverters of the proposed architecture, which have no explicit load element to modulate; introducing one would add capacitance to the only high-speed node of each slicer and degrade its switching delay. Dynamic offset-cancellation schemes such as auto-zeroing were excluded for the reasons detailed in Section 2.6.4: they require periodic clock phases, which would reintroduce into the converter precisely the clocked activity that the asynchronous architecture is designed to eliminate.

Bulk-driven adjustment, by contrast, satisfies all the constraints of the event-driven operation simultaneously. The bulk terminals of the slicer transistors are not part of the signal path, so driving them from a static bias source adds no capacitance to the switching nodes and consumes

no dynamic power. Once the correction voltages are set, they are held by purely DC conduction paths, preserving the idle behaviour of the converter when the input is inactive. Finally, because the NMOS and PMOS bulk potentials act on the pull-down and pull-up networks independently, they provide exactly the two degrees of freedom needed to counteract both the symmetric (SS, FF) and the asymmetric (FS, SF) corner perturbations characterised in Section 4.3.

4.2 Function of NMOS and PMOS Transistors Under Different Body Bias

The threshold voltage of a MOSFET is not solely determined by the device geometry and doping profile but is also a function of the potential applied to its body terminal relative to the source. This dependence, commonly referred to as the body effect or substrate bias effect, is described for an NMOS transistor by

$$V_{tn} = V_{tn0} + \gamma_n \left(\sqrt{2\phi_{F,n} + V_{SB}} - \sqrt{2\phi_{F,n}} \right), \quad (4.1)$$

where V_{tn0} is the zero-bias threshold voltage, γ_n is the body effect coefficient, $\phi_{F,n}$ is the Fermi potential, and V_{SB} is the source-to-body voltage. The analogous expression for a PMOS transistor, accounting for the sign conventions appropriate to a p-channel device, takes the form

$$|V_{tp}| = |V_{tp0}| + \gamma_p \left(\sqrt{2\phi_{F,p} + |V_{SB}|} - \sqrt{2\phi_{F,p}} \right), \quad (4.2)$$

where the relevant source-to-body voltage for the PMOS is determined by the potential difference between the source and the N-well. In both cases, an increase in $|V_{SB}|$ raises the magnitude of the threshold voltage, a condition referred to as reverse body biasing (RBB). Conversely, reducing $|V_{SB}|$ below its nominal value or, in technologies that permit it, forward-biasing the body junction lowers $|V_t|$, a condition known as forward body biasing (FBB).

A direct consequence for the slicer inverters in the proposed architecture is that each cell's trip point, given by equation (3.4), depends on the threshold voltages of both transistors composing the inverter. By substituting the body-bias-dependent expressions for V_{tn} and $|V_{tp}|$ into the inverter threshold equation and expanding to first order, it is apparent that a positive shift in V_{tn} achieved by applying RBB to the NMOS weakens the pull-down network relative to the pull-up network and therefore raises V_M , while an equivalent increase in $|V_{tp}|$ weakens the pull-up network and lowers V_M . This asymmetry between the NMOS and PMOS body bias responses provides two independent degrees of freedom for adjusting the trip point of each slicer: the NMOS bulk voltage controls the effective strength of the pull-down path, while the PMOS bulk voltage (equivalently, the N-well potential) controls the pull-up path. Applying corrections simultaneously to both body terminals therefore allows independent tuning of the trip point magnitude and of the symmetry of the transition characteristic, providing a richer correction space than either terminal alone could offer.

4.3 Process Corner Dependence and Motivation for Trimming

In a deep-submicron CMOS process, fabrication variability causes the threshold voltages and carrier mobilities of the NMOS and PMOS devices to deviate from their nominal (typical-typical, TT) values. The standard process corners used to bound this variability, namely slow-slow (SS), fast-fast (FF), fast-NMOS slow-PMOS (FS) and slow-NMOS fast-PMOS (SF), capture the four extreme combinations of NMOS and PMOS behaviour and represent the conditions under which circuit performance is most likely to degrade.

For the slicer-based thermometer coder described in Chapter 3, the impact of process variation is particularly significant because the thermometer code relies on the precise relative ordering of the inverter trip points across the output bank. At the typical corner, the ratios r_k assigned to each slicer place the trip points at the intended positions relative to the reference level of 0.4 V, and the resulting differential non-linearity (DNL) and integral non-linearity (INL) of the converter remain within specification. At non-typical corners, however, the effective transistor strengths deviate from their nominal values in a manner that is not uniform across the bank: since the PMOS and NMOS devices within each slicer are affected differently depending on the corner, the trip point of each inverter shifts by a cell-dependent amount, breaking the intended monotonic spacing and introducing non-linearities that manifest as elevated DNL and INL.

At the SS corner, both NMOS and PMOS threshold voltages increase, reducing the drive strength of both pull-down and pull-up networks. Although the symmetric nature of this degradation might suggest a self-compensating effect on the trip point, the body effect coefficients γ_n and γ_p and the nominal threshold values are in general not matched between the two device types, so the net shift in V_M is non-zero and, more importantly, cell-dependent due to the different sizing ratios across the slicer bank. The FF corner produces the complementary effect: reduced threshold voltages increase the drive strength of both device types, again shifting the trip points by amounts that vary across the bank. The FS and SF corners are qualitatively different in that they introduce an asymmetric perturbation. At FS the NMOS is faster than nominal while the PMOS is slower, which biases all slicers toward a lower trip point and compresses the effective decision levels toward the bottom of the input range; at SF the complementary imbalance expands the decision levels toward the top. In all four non-typical corners, the loss of linearity was found to cause DNL excursions beyond 0.5 LSB, which takes the converter outside its linearity target under these conditions.

4.4 Offline Trimming Strategy via Body Bias Adjustment

To restore the linearity of the thermometer coder across process corners without modifying the core circuit topology, an offline trimming strategy based on body bias adjustment was adopted. The principle of the approach is to apply distinct bulk voltages to the NMOS and PMOS transistors of the output slicer stage independently for each device type in a manner calibrated to shift the trip points back toward their nominal positions. Since both the NMOS bulk voltage V_{BN} and the

PMOS bulk voltage V_{BP} contribute independently to the trip point through equations (4.1) and (4.2), respectively, adjusting both simultaneously corrects both the common-mode shift of the trip points, which affects the DNL, and the differential imbalance between NMOS-dominated and PMOS-dominated corners, which affects the INL. The trimming is offline in the sense that the bulk voltages are determined once, from the identified process corner, and held constant during normal operation; no continuous adaptive feedback is involved.

4.4.1 Determination of the Correction Pairs

For each of the four non-typical corners, a specific pair of values (V_{BN} , V_{BP}) was determined with the objective of bringing the DNL of the converted thermometer code below 0.5 LSB across the full input range; the degree to which each corner meets that target is reported in Chapter 6. The pairs were obtained by sweeping the two bulk voltages over the range permitted by the technology, extracting the trip points of all slicers at each bias combination, and selecting the combination that minimises the worst-case deviation of the measured trip points from their nominal TT positions. At the SS corner, where both device types exhibit elevated threshold voltages and therefore reduced drive strength, forward body biasing is applied to both the NMOS and PMOS, reducing V_{tn} and $|V_{tp}|$ toward their nominal values and thereby restoring the intended trip point positions. At the FF corner, reverse body biasing is applied to both types to counteract the excessive drive strength resulting from the low threshold voltages. The FS and SF corners require asymmetric corrections. At FS, a stronger reverse bias on the NMOS bulk weakens the pull-down network to compensate for the fast-NMOS tendency, while the PMOS bulk is forward-biased to recover part of the lost pull-up strength; the SF corner is handled symmetrically, with the dominant correction applied to the PMOS bulk. The resulting correction table is summarised in Table 4.1.

Table 4.1: Body bias correction vectors determined for each process corner.

Corner	V_{BN} (V)	V_{BP} (V)	Bias type (NMOS / PMOS)
TT			nominal / nominal
SS			FBB / FBB
FF			RBB / RBB
FS			RBB / FBB
SF			FBB / RBB

4.4.2 Verilog-A Behavioural Implementation

The trimming was implemented in simulation using a Verilog-A behavioural model. The model monitors the process corner identifier and, based on the detected corner, drives V_{BN} and V_{BP} to the pre-computed correction values of Table 4.1. The two bias voltages are exposed as electrical outputs of the behavioural block and routed to the bulk terminals of the NMOS devices and to

the N-well contacts of the PMOS devices of the slicer bank, so that the correction acts on the transistor-level netlist without requiring any modification to it. This approach allows the trimming to be evaluated across all corners within the same simulation framework as the nominal design. The Verilog-A abstraction also provides a clean interface for extending the trimming table to intermediate corner points or to temperature-corner combinations should a finer correction granularity be required in future iterations. The effect of the body bias corrections on the DNL and INL of the converter is presented in Chapter 6.

Chapter 5

Synchronous Reference Flash ADC

5.1 Motivation for Architectural Selection

Benchmarking a novel converter architecture requires a reference point that is close enough structurally to make the comparison genuinely informative. The synchronous reference used in this work is a 4-bit Flash ADC with rail-to-rail comparators, originally presented in [34] and designed in a 65-nm CMOS process. For the purpose of this comparison, the architecture was re-implemented and simulated in a 16-nm technology node, under the same conditions applied to the proposed asynchronous converter throughout this dissertation.

The choice of this particular design was not arbitrary. Several properties directly align with those of the asynchronous architecture described in Chapter 3, and it is precisely this alignment that makes the comparison meaningful. Both converters use the Flash topology, in which the input is compared in parallel against a set of fixed decision thresholds and the result is produced as a thermometer code, so the fundamental conversion principle is shared. Both are designed for 4-bit resolution with $2^4 - 1 = 15$ parallel comparison elements, so the complexity of the comparison stage is the same on both sides. Both were implemented in the same 16-nm technology node and operate from the same 0.8 V supply, eliminating technology and supply voltage as sources of asymmetry. Finally, both architectures are explicitly oriented towards low-power operation, which is the design space where the asynchronous approach is expected to have the most significant impact.

One difference between the designs must nevertheless be stated explicitly so that the comparison remains honest: the input ranges are close but not identical. The StrongARM comparators of the reference resolve inputs all the way to the rails, while the complementary follower of the asynchronous front-end covers almost the entire supply range but loses functionality within a small margin of each rail, as documented in Chapter 3. This asymmetry is accounted for by characterising both converters over their respective effective input ranges, and the energy comparison is in any case performed through resolution-normalised figures of merit, in which the effective number of bits of each converter appears explicitly.

The fundamental difference, and the one this comparison is intended to illuminate, is the presence of a global clock. The synchronous reference depends on a periodic clock to coordinate its comparators and define valid output instants, while the asynchronous architecture eliminates the clock entirely. Removing the clock distribution network is one of the primary motivations for adopting asynchronous operation, since the clock tree contributes a non-trivial fraction of total power in high-speed designs. By holding the remaining parameters constant and changing only this aspect, the comparison gives a direct, controlled view of what asynchronous operation actually costs, or saves, in power and performance.

5.2 Architecture Overview

A Flash ADC resolves an analogue input voltage in a single conversion step by comparing it simultaneously against $2^N - 1$ evenly spaced reference levels, where N is the number of output bits. For a 4-bit converter, this means 15 comparators operate in parallel at every active clock edge, each producing a binary decision about whether the input exceeds its assigned threshold. The outputs of all 15 comparators form a thermometer code (a sequence of ones followed by zeros, with no transition other than the boundary between them), which an encoder then maps to the final 4-bit binary word.

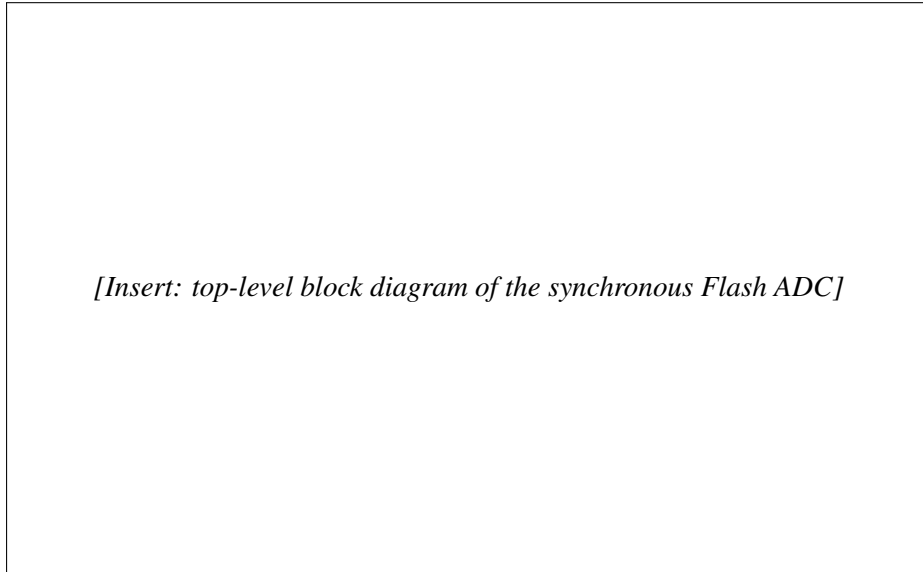


Figure 5.1: Top-level block diagram of the synchronous 4-bit Flash ADC [34].

The reference voltages fed to each comparator are generated by a resistor ladder connected between V_{REF+} and V_{REF-} , which in this rail-to-rail implementation are tied directly to the supply rails V_{DD} and V_{SS} . The ladder divides the full input range into 15 equally spaced steps of one LSB each, so the k -th comparator from the bottom receives a reference of

$$V_{REF,k} = V_{SS} + k \cdot \frac{V_{DD} - V_{SS}}{2^N}, \quad k = 1, 2, \dots, 2^N - 1. \quad (5.1)$$

Since all comparators are driven by the same clock, every decision is made within a single clock period and the output is valid only at well-defined time instants. This is the defining characteristic of a synchronous converter and stands in direct contrast to the asynchronous operation discussed in Chapter 3.

5.3 The StrongARM Latch Comparator

5.3.1 Circuit Description and Operating Principle

The comparator used in this architecture is the StrongARM latch, a dynamic topology that has become something of a standard reference in both high-speed receiver design and data converter circuits [112]. It is widely adopted for good reasons: high regeneration speed, good noise performance and essentially zero static power dissipation, all of which matter in a Flash ADC where 15 comparators operate in parallel.

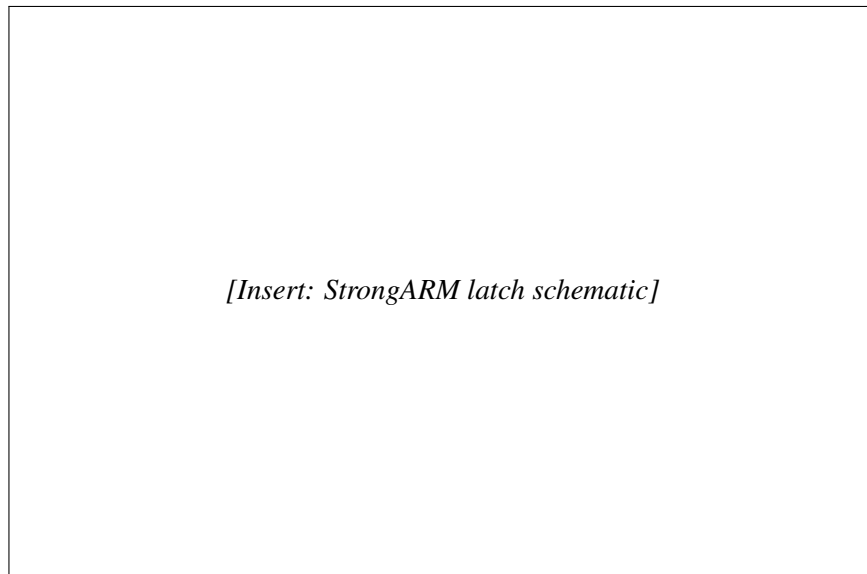


Figure 5.2: Schematic of the StrongARM latch comparator [34].

The circuit operates in two alternating phases controlled by the clock signal: a reset phase and an evaluation phase.

The reset phase occurs when the clock is low, the tail transistor is turned off and a pair of reset switches pull both output nodes up to V_{DD} . This precharges the outputs to a known state before every comparison, so that each evaluation begins from the same initial condition regardless of the previous result. Because the tail transistor is off and no current path exists through the core of the circuit, static power dissipation is identically zero during this phase.

The evaluation phase occurs when the clock rises, the tail transistor turns on and current begins to flow through the differential input pair. The input transistors steer this current according to the voltage difference $\Delta V = V_{INP} - V_{INN}$: the side with the higher input voltage conducts more, causing that output node to discharge faster than the other. Once a sufficient imbalance develops across

the internal nodes, the cross-coupled inverter pair at the output enters regeneration, a positive feedback mechanism that amplifies the initial imbalance exponentially until the outputs settle at valid logic levels [112]. The time needed to complete the regeneration is inversely related to the input overdrive: larger differences between the input voltage and the reference threshold resolve faster, while inputs very close to the threshold require more time. This input-dependent latency is a fundamental property of dynamic comparators and is one of the key differences exploited by the asynchronous architecture.

Since current only flows during the evaluation phase, and only briefly at that, the average power consumption is directly proportional to the clock frequency. This makes the StrongARM latch well suited to low-power converter design: reducing the sampling rate reduces the power linearly.

5.3.2 Rail-to-Rail Input Range

A key property of this comparator implementation is its ability to correctly resolve input voltages across the full supply range, from V_{SS} to V_{DD} . In a Flash ADC, the comparators assigned to the top and bottom reference levels must operate very close to the supply rails, and a comparator that loses sensitivity or gain in those regions would introduce nonlinearity at the extremes of the transfer characteristic, degrading the integral linearity of the converter. The rail-to-rail design ensures this does not happen, maintaining consistent performance across the entire input range.

The asynchronous front-end pursues the same goal with a complementary input stage, but, as documented in Chapter 3, its effective window stops slightly short of both rails due to the overdrive lost by the input pairs near the supply limits. The benchmarking methodology described in Chapter 6 therefore evaluates each converter over its own effective input range.

5.4 Clock Generation

Since the architecture is synchronous, a stable clock signal must drive the comparator array and determine the conversion instants. For simulation purposes, the clock was generated using a ring oscillator, which provides a compact and realistic on-chip clock source without requiring an ideal external generator.

Rather than driving the comparators directly from the oscillator output, the signal was routed through a tapered inverter buffer chain designed to model the loading and parasitics of a realistic clock distribution network. The chain consists of a sequence of inverters with progressively increasing drive strength, with each stage sized at approximately twice the fanout of the previous one. The final stage of the chain is a $\times 24$ fanout inverter, whose size reflects the capacitive load that a clock signal meets when distributed through the metal routing to the ADC core: the interconnect wires, vias and comparator gate inputs that the clock must charge and discharge on every cycle.

A $\times 24$ final buffer is a reasonable and representative choice for modelling the routing from a clock source to the ADC, as it captures the kind of loading one would expect in a physically

compact but realistic clock distribution. Neglecting this stage would result in a power estimate that is overly optimistic: in a real implementation, the clock tree is rarely a negligible contributor to total power, and any fair assessment of the synchronous architecture must account for it.

The complete clock path used in simulation is therefore the ring oscillator, followed by the buffer chain ($\times 2$ fanout per stage), the $\times 24$ output buffer and, finally, the comparator array.

The power measurement methodology applied to this reference converter, together with its simulated performance figures, is presented alongside the results of the proposed asynchronous design in Chapter 6, so that the two converters are characterised and compared within a single framework.

Chapter 6

Simulation Results and Performance Benchmarking

6.1 Simulation Setup and Input Test Signals

To validate the theoretical framework of Chapter 3 and evaluate the physical performance of the proposed asynchronous Flash ADC, transistor-level simulations were run in a 16-nm CMOS process at a nominal supply of 0.8 V. This section describes the simulation environment, the reconstruction methodology required for non-uniform asynchronous data, and the input test stimuli.

Two families of stimuli were used. Static linearity was measured with a trapezoidal pulse spanning the effective input range of 0.1 V to 0.7 V (Section 3.5.1), with equal rise and fall ramps swept from 35 ns down to 5 ns and a fixed 20 ns hold between them. Dynamic performance was measured with a full-scale sinusoid at $f_{in} = 9$ MHz (amplitude 0.3 V around a 0.4 V common mode, again spanning 0.1–0.7 V), simulated for 200 μ s with transient noise enabled.

In a synchronous converter the output samples sit on a periodic clock grid. The asynchronous level-crossing Flash ADC instead produces output transitions only when a quantisation threshold is crossed, so a standard Fast Fourier Transform (FFT), which requires uniformly sampled records, cannot be applied directly to the raw output. The reconstruction used here is deliberately simple: the fifteen digital outputs and the input are linearly interpolated onto a uniform analysis grid of 180 MHz, each interpolated output is compared against the logic threshold of 0.4 V to recover the thermometer bits, the output code is the count of asserted bits, and the equivalent analog value follows from mapping that code back onto the 0.1–0.7 V range. The analysis grid is twenty times the input frequency and coherent with it ($N = 36000$ samples place the fundamental exactly on bin 1800), so the FFT uses a rectangular window and no spectral leakage correction is needed. The Signal-to-Noise Ratio (SNR), the Signal-to-Noise-and-Distortion Ratio (SINAD), the Effective Number of Bits (ENOB), the Total Harmonic Distortion (THD, first seven harmonics, aliased into the first Nyquist zone where necessary) and the Spurious-Free Dynamic Range (SFDR) are extracted from this spectrum.

The same resampled record defines the bubble error rate used throughout this chapter: a sample exhibits a bubble error when its fifteen-bit thermometer pattern contains more than one $0 \leftrightarrow 1$ transition, and the bubble rate is the fraction of analysis samples for which this happens.

Figure 6.1: Input sinusoid and reconstructed converter output over the first microsecond of the dynamic test.

Figure 6.1 shows the input sinusoid against the reconstructed output over the first microsecond of the test, and Figure 6.2 shows the spectrum of the reconstructed output, with the fundamental signal bin, the noise floor and the harmonic distortion components.

Figure 6.2: FFT spectrum of the reconstructed asynchronous Flash ADC output signal.

Figure 6.3 maps the SNDR and the ENOB over a broad range of input frequencies and amplitudes. The curves contrast the ideal noiseless performance with the degraded values under transient noise, and record the incidence of noise-induced threshold chatter.

The choice of 9 MHz is tied to the hysteresis analysis of Section 6.2.1. A full-scale sinusoid at f_{in} has a peak slew rate of $2\pi f_{in}A$, which for 9 MHz and 0.3 V of amplitude gives ≈ 17.0 V/ μ s, the same slew rate as the fastest ramp that meets the 1 LSB hysteresis criterion (0.6 V/35 ns ≈ 17.1 V/ μ s). The sinusoid therefore never exceeds the tracking envelope validated by the ramp test, and the hysteresis observed at faster ramps does not affect the dynamic measurement; the absence of bubble errors over the whole record confirms it. By the same criterion, a full-scale sinusoid at 20 MHz would reach 37.7 V/ μ s and exceed the envelope, so 9 MHz is, in practice, the full-scale sinusoidal bandwidth of the converter.

Figure 6.3: SNDR and ENOB as a function of the input signal frequency and amplitude.

Table 6.1 summarises the dynamic metrics extracted from the 9 MHz transient noise test. The SINAD of 26.50 dB corresponds to an ENOB of 4.11 bits, marginally above the nominal 4-bit resolution; this is consistent with the level-crossing property discussed in Chapter 2, whereby the absence of a uniform sampling clock removes the conventional quantisation noise floor and can push the SNDR past the limit of a clocked converter with the same nominal resolution. No bubble errors were observed over the 36000 analysis samples of this test.

Table 6.1: Dynamic performance of the asynchronous converter ($f_{in} = 9$ MHz, full-scale input, with transient noise).

Metric	Value	Unit
SINAD	26.50	dB
ENOB	4.11	bit
SNR	31.55	dB
THD	−28.13	dB
SFDR	30.63	dB
Bubble rate	0.00	%

6.2 Static Linearity Results

The static linearity metrics, Differential Non-Linearity (DNL) and Integral Non-Linearity (INL), quantify the deterministic mismatches of the architecture and the circuit. The proposed asynchronous Flash ADC contains no reference ladder, so its DNL and INL are set by the deviations of the slicer trip points from their nominal sizing-defined positions: the deterministic, corner-dependent shifts analysed in Chapter 4, plus the random threshold variations of the slicer bank.

6.2.1 Threshold Extraction and Linearity Versus Conversion Rate

The fifteen decision thresholds were measured with the trapezoidal pulse described in Section 6.1. For each slicer, the input voltage at which its digital output crosses the 0.4 V logic threshold is recorded by linear interpolation, once on the rising ramp and once on the falling ramp. The difference between the two readings is the input-referred hysteresis of that slicer; their average is taken as its decision threshold. The DNL of each of the sixteen codes follows from the widths between consecutive sorted thresholds relative to the ideal LSB of 37.5 mV, the INL of each transition from the deviation of its threshold from the ideal uniform level, and the offset from the mean of those deviations. Since the widths of the two end codes depend directly on the chosen range limits, the DNL figures reported below exclude them and cover codes 1 to 14.

The hysteresis is not a designed property of the slicers: it is the input-referred signature of the conversion delay τ_{conv} modelled in Section 3.6.1. While the input ramps, every delay in the follower and the slicer translates into a voltage error proportional to the ramp rate, so the rising and falling readings separate as the ramp gets steeper. A measurement is considered valid when the worst-case hysteresis stays below 1 LSB. Table 6.2 reports the static metrics for ramp times from 35 ns down to 5 ns, with the equivalent conversion rate computed from the active ramp time (and, in parentheses in the discussion below, with the 20 ns hold included).

Table 6.2: Static linearity and hysteresis of the asynchronous converter versus input ramp time. DNL excludes the two end codes. Valid means worst-case hysteresis below 1 LSB.

Ramp (ns)	Rate (MHz)	Offset (LSB)	Max DNL (LSB)	Max INL (LSB)	Max hyst. (LSB)	Valid
35	14.3	−0.012	0.412	0.518	0.959	Yes
30	16.7	0.019	0.403	0.486	1.049	No
25	20.0	0.010	0.315	0.453	1.163	No
10	50.0	−0.046	0.456	0.351	2.494	No
5	100.0	−0.154	0.350	0.937	4.873	No

The data confirms the behaviour predicted by the slew rate model. The DNL, INL and offset stay essentially flat across the sweep (all fifteen thresholds were found in every run, with no missing codes), but the hysteresis grows roughly in proportion to the ramp rate, from 0.96 LSB at 35 ns to 4.87 LSB at 5 ns. The hysteresis is therefore the binding constraint on speed: the fastest ramp that meets the 1 LSB criterion is 35 ns, which corresponds to a maximum conversion rate of 14.3 MHz over the active ramps (11.1 MHz when the 20 ns hold is included). Figure 6.4 plots the three error metrics against the conversion rate.

Figure 6.4: Worst-case DNL, INL and hysteresis as a function of the equivalent conversion rate.

6.2.2 Linearity Across Process Corners and Trimming

These non-idealities are corrected by the body bias trimming technique implemented through the Verilog-A behavioural framework of Chapter 4. To measure its effect, the static error profiles were extracted before and after applying the trimming vectors at each process corner, separately on the rising and on the falling ramp.

At the typical corner no correction is needed: both the DNL and the INL stay within approximately ± 0.15 LSB across all codes (Figure 6.5). Table 6.3 reports the worst-case errors at the four non-typical corners, before and after trimming.

Figure 6.5: DNL per code and INL per transition of the asynchronous converter at the typical (TT) corner.

The correction is effective but not uniform across corners. The worst-case DNL improves at all four corners, falling below the 0.5 LSB target at SS, FF and SF and remaining marginally above it at FS (0.586 LSB on the falling ramp). The INL gains are largest where the untrimmed errors were worst: at SS the falling-ramp INL drops from 1.09 to 0.02 LSB, and at FF from 1.37 to 0.35 LSB. The SF corner is the exception: its body bias pair trades INL for DNL, so while the DNL improves, the worst-case INL degrades from 1.08 to 1.43 LSB. A single global (V_{BN}, V_{BP}) pair per corner

Table 6.3: Worst-case static linearity per process corner, before and after body bias trimming, measured on the rising and falling ramps (values in LSB).

Corner	DNL rise		DNL fall		INL rise		INL fall	
	Before	After	Before	After	Before	After	Before	After
SS	0.6958	0.4882	0.6628	0.5043	0.9489	0.1905	1.0949	0.0213
FF	0.3814	0.4008	0.6372	0.3636	0.6670	0.7435	1.3724	0.3514
FS	0.3814	0.4979	0.6372	0.5859	0.6670	0.6614	1.3724	0.9807
SF	0.5191	0.3414	0.5142	0.4700	0.5829	1.0261	1.0787	1.4333

cannot optimise both metrics at once when the perturbation is asymmetric; per-slicer calibration, discussed in Chapter 7, would remove that constraint.

6.3 Random Mismatch: Monte Carlo Analysis

The corner-indexed trimming of Chapter 4 corrects deterministic shifts only; random intra-die mismatch remains uncorrected by design. To quantify how much linearity that limitation costs, a 330-run Monte Carlo simulation was performed on two adjacent slicer outputs, out_0 and out_1 . In each run, the decision threshold of each output was extracted as the input voltage at which the output crosses the 0.4 V logic level, and the width of the code between them follows from the difference of the two thresholds; the DNL of that code is the deviation of this width from the ideal LSB. Of the 330 runs, 106 produced a measurable crossing for both outputs; the statistics below are computed over those valid runs.

Table 6.4: Monte Carlo statistics of the threshold spacing between two adjacent slicers (out_0 , out_1), without trimming.

Quantity	Value	Unit
Monte Carlo runs (total / valid)	330 / 106	–
Code width, standard deviation	215.6	mV
DNL of the code, mean	−2.02	LSB
DNL of the code, standard deviation	5.75	LSB
Worst DNL observed	15.18	LSB
Runs free of missing code	42.5	%
Runs with $ DNL \leq 0.5$ LSB	7.6	%
Runs with $ DNL \leq 1.0$ LSB	16.0	%

The standard deviation of the code width, 215.6 mV or 5.75 LSB, is almost six times the ideal LSB of 37.5 mV. Fewer than half of the valid runs avoid a missing code between the two outputs, and fewer than one run in ten keeps the DNL of this single code within ± 0.5 LSB. Figure 6.6

shows the distribution. Random threshold mismatch, rather than corner variation, is therefore the dominant linearity hazard of the slicer bank. This confirms the limitation acknowledged in Chapter 7 and puts numbers behind the per-slicer foreground calibration proposed there as future work.

Figure 6.6: Monte Carlo distribution of the DNL of the code between two adjacent slicers, with the ± 0.5 LSB limits and the missing-code boundary marked.

6.4 Power Dissipation and Energy Efficiency

A defining feature of the asynchronous Flash ADC is its signal-dependent power profile. In direct contrast to traditional synchronous architectures that exhibit a static power baseline due to continuous clock toggling, the dynamic power of the proposed converter is governed strictly by the frequency of threshold-crossing events.

6.4.1 Power Measurement Methodology

To keep the comparison between the two converters fair, the power figures reported here cover only the core conversion path of each design. The encoder, that is, the digital logic that translates the thermometer code into the final binary word, was excluded from all power measurements on both sides of the comparison.

The reasoning is straightforward: both designs require some form of thermometer-to-binary conversion at their outputs, and this encoding logic is largely independent of whether the preceding conversion is synchronous or asynchronous. Including it on one side of the comparison but not the other would artificially inflate one power budget and lead to a conclusion that does not fairly reflect the underlying architectural trade-off. The most defensible approach is therefore to exclude it from both and compare only the converter cores.

Under this methodology, the power reported for the synchronous reference in Table 6.7 includes the resistor ladder, the 15 StrongARM latch comparators, the ring oscillator and the tapered clock buffer chain described in Chapter 5. The clock distribution path is precisely the overhead that

the asynchronous design eliminates. The power reported for the proposed asynchronous converter includes the voltage follower, the slicer bank and the event-generation logic of Chapter 3.

6.4.2 Signal-Dependent Power Scaling

Two operating points were measured directly. With the input active, the converter draws 181 μW ; with the input idle, it draws 145 μW . The idle figure is the static floor anticipated in Section 3.6.2: because the complementary follower operates within the 0.1–0.7 V window and the input never reaches the bottom rail, the input stages conduct a residual bias current even when nothing is being converted. The difference between the two measurements, 36 μW , is the activity-dependent component of the consumption. Figure 6.7 shows the dependency of the total power on the input frequency.

Figure 6.7: Total and dynamic power consumption as a function of the input signal frequency (f_{in}).

Energy efficiency in this non-uniform context is assessed with the adapted Walden Figure of Merit ($\text{FoM}_{W,adapted}$) introduced in Chapter 2, which uses the effective average sampling rate ($f_{s,avg}$) set by the crossing density and therefore allows a fair benchmark against synchronous designs:

$$\text{FoM}_{W,adapted} = \frac{P_{total}}{2^{\text{ENOB}} \cdot f_{s,avg}} \quad (6.1)$$

In this work, $f_{s,avg}$ is obtained directly from the simulation record: the conversion events generated over the observation window are counted and divided by the window duration. Table 6.8 reports each figure of merit next to the $f_{s,avg}$ used to compute it.

The measured power figures are summarised in Table 6.5.

6.5 Synchronous Reference Flash ADC Results

The synchronous reference converter described in Chapter 5 was simulated under the same 16-nm, 0.8 V conditions as the proposed design, following the power measurement methodology of Section 6.4.1. Its static linearity was characterised with a 20 MHz input sampled at 640 MS/s over the same 0.1–0.7 V range. The power was measured at the workload-matched operating point

Table 6.5: Measured power consumption and energy efficiency of the asynchronous converter.

Performance Parameter	Value	Unit
Power consumption, input active	181	μW
Power consumption, input idle	145	μW
Activity-dependent component	36	μW
FoM _W	fJ/conv.-step	

defined below, a 9 MHz full-scale input sampled at 288 MS/s, and the consumption came out the same, 727 μW , at both clock rates.

Neither operating point comes from the original publication. The design of [34] was demonstrated at 20 MS/s with a 332 kHz input in 65 nm; reproducing that point here would compare the two converters in a regime unrelated to the one the asynchronous design targets. The matched point follows instead from the activity model of Section 3.6.2 and from the slew rate envelope of Section 6.1: the fastest full-scale sinusoid the asynchronous converter tracks cleanly is 9 MHz, and at that input a 4-bit level-crossing converter generates $f_{s,avg} = 2^{N+1} f_{in} = 32 \times 9 \text{ MHz} = 288 \text{ M}$ crossings per second. Clocked at 288 MS/s with the same 9 MHz stimulus, the synchronous reference performs the same number of conversions per second, on the same input, as the asynchronous converter in the test of Section 6.1. The power figures of the two converters therefore describe the same workload. Its static linearity is good: the worst-case DNL and INL both stay within approximately 0.27 LSB, and the transfer function shows all sixteen codes with no missing levels (Figure 6.8). The performance summary is presented in Table 6.7 and serves as the baseline for the comparative analysis of the following section.

Figure 6.8: Static characterisation of the synchronous reference: DNL and INL versus input voltage, and measured transfer function against the ideal staircase (20 MHz input, 640 MHz clock).

The power consumption of the synchronous reference is not a single block. In any clocked Flash converter the clock must physically reach all fifteen comparators, so a converter of this kind always carries a clock generator and a distribution network between the generator and the array.

That routing belongs to the architecture itself, and any honest power figure for a synchronous Flash has to include it. As described in Chapter 5, the distribution was modelled with a tapered buffer chain whose final $\times 24$ fanout stage represents the load of a normal, physically compact route to the comparator bank. Table 6.6 separates the measured consumption into the converter core, the ring oscillator and the clock routing; the complete set draws 727 μW .

Table 6.6: Power breakdown of the synchronous reference Flash ADC.

Block	Power	Unit
ADC core (comparators and reference ladder)		μW
Ring oscillator		μW
Clock routing (buffer chain, $\times 24$ fanout)		μW
Complete converter	727	μW

6.6 Comparative Analysis and Discussion

Table 6.8 puts the developed design in context. It compares the core performance metrics of the synchronous reference Flash ADC of Chapter 5, the proposed asynchronous Flash ADC, and recent state-of-the-art architectures from the literature.

The comparison makes the trade-off concrete. On the same 9 MHz full-scale stimulus and at a matched conversions-per-second workload (Section 6.5), the complete synchronous converter draws 727 μW against 181 μW for the active asynchronous core, a factor of four. A substantial part of the synchronous budget goes to clock generation and distribution, blocks that do not exist in the asynchronous design. The synchronous baseline keeps the simpler control block and the higher conversion rate, but its power envelope is rigid and its efficiency collapses when the input is barely active. The asynchronous implementation scales its energy with the input activity and reaches a competitive Walden Figure of Merit, particularly once the body bias trimming network restores the effective resolution under process variations.

A note on the fairness of this comparison is warranted, since an activity-indexed figure of merit could be suspected of favouring the asynchronous design by construction. Three observations bound that concern. First, both converters have the same 4-bit resolution and their cores are measured under the same exclusions (Section 6.4.1), so the structural terms of the comparison are symmetric. Second, the advantage predicted by the activity-dependent model is conditional, and the condition is stated by the model itself (Section 3.6.2): as input activity rises, $f_{s,avg}$ approaches $2^{N+1}f_{in}$ and the dynamic power of the asynchronous core grows linearly, until near the slope-overload limit the activity advantage disappears altogether. The asynchronous architecture is expected to win for sparse inputs and to lose its margin for continuously active inputs near full bandwidth; this comparison targets the former regime because that is the application space defined

Table 6.7: Simulated performance summary of the synchronous reference Flash ADC in 16-nm CMOS.

Parameter	Value	Unit
Supply Voltage	0.8	V
Resolution	4	bit
Sampling Frequency	288	MS/s
Power Consumption	727	μ W
Rail-to-Rail Input	Yes	–
Bubble Rate		%
FoM _W (Walden)		fJ/conv.-step
<i>Dynamic Performance (without Noise)</i>		
SNR		dB
SINAD / SNDR		dB
ENOB		bit
<i>Dynamic Performance (with Transient Noise)</i>		
SNR		dB
SINAD / SNDR		dB
ENOB		bit

Table 6.8: Performance comparison with the synchronous baseline and state-of-the-art literature.

Parameter	Synchronous Baseline (Chapter 5)	This Work (Asynchronous)	Ref. [X]	Ref. [Y]	Ref. [Z]
Technology (nm)	16	16			
Supply Voltage (V)	0.8	0.8			
Resolution (Bits)	4	4			
f_s or $f_{s,avg}$ (MS/s)	288				
SNDR (dB)		26.50			
ENOB (Bits)		4.11			
Power (μ W)	727	181			
FoM _W (fJ/c.-step)					
Architecture	Synchronous Flash	Asynchronous Flash			

in Chapter 1. Third, the one structural asymmetry that remains, the restricted effective input range of the complementary follower, runs against the asynchronous design rather than in its favour.

Chapter 7

Conclusions and Suggestions for Future Work

7.1 Summary of the Work Developed

This dissertation addressed the design and validation of an asynchronous Flash Analog-to-Digital Converter intended for low-power, event-driven applications such as Internet of Things sensor nodes and biomedical signal acquisition interfaces. The central motivation for this work stems from the fundamental inefficiency of conventional synchronous ADC architectures when dealing with time-sparse signals: a global clock forces continuous switching activity across the comparator bank and clock distribution network, dissipating dynamic power regardless of whether the input signal is actually carrying new information. Removing that clock and replacing it with a level-crossing, event-driven mechanism was the core hypothesis explored throughout the dissertation.

The work began with a review of the state of the art in data conversion. The classical ADC architectures (Flash, SAR, Pipeline and Sigma-Delta) were examined with particular attention to the trade-offs between speed, resolution and power consumption. This analysis was grounded in performance data from 140 publications drawn from ISSCC and JSSC, from which Walden and Schreier figures of merit were extracted and compared across a wide range of sampling frequencies and technology nodes. The review made clear that synchronous Flash converters occupy the high-speed frontier but at the cost of a poor energy figure of merit, and that asynchronous, event-driven operation is one of the most promising paths toward breaking this constraint for sparse-signal applications. A dedicated discussion on the adaptation of standard figures of merit to asynchronous systems was also included, since applying a fixed Nyquist sampling frequency to a level-crossing converter is inherently inappropriate and leads to misleading efficiency estimates.

The design of the asynchronous ADC itself evolved through three distinct architectural phases. The first candidate was a current-mirror flash configuration in which a transconductance cell converts the differential input voltage into an error current, and a current-to-code converter maps that current to a thermometer-coded output by exploiting asymmetric PMOS and NMOS mirror ratios at each comparison node. This topology operates in continuous time and requires no

external clock, which made it a natural fit for the event-driven approach. The second phase investigated a complementary dual-input architecture aimed at extending the input common-mode range toward rail-to-rail operation by combining an NMOS and a PMOS differential pair with cross-coupled output branches. Although conceptually attractive, this implementation could not be made to work within the 0.8 V supply budget of the target 16 nm process: the mirror loads and the cross-coupling stacked additional transistors in every signal path, the accumulated overdrive requirements exceeded the available voltage headroom, and no feasible bias point could be found that kept all devices in saturation simultaneously. The third and final architecture kept the complementary rail-to-rail input but rebuilt it with two five-transistor OTAs, one with an NMOS and one with a PMOS input pair, each loaded by a single active current mirror and biased through the low-voltage tail mirror retained from the second phase. With far fewer stacked devices, this arrangement fits the available headroom, and the complementary OTA, configured as a unity-gain voltage follower, drives a bank of CMOS slicer inverters through its low-impedance output. Each slicer is sized with a distinct PMOS-to-NMOS transistor ratio, which sets its switching threshold at a specific voltage level within the supply range. The ensemble of slicers produces an asynchronous thermometer code whose transitions directly encode level-crossing events, with no clock required at any point in the signal path.

Alongside the asynchronous core, a synchronous reference Flash ADC was implemented in the same 16 nm technology and under the same supply conditions to serve as a fair benchmark. This reference uses fifteen StrongARM latch comparators driven by a ring oscillator with a tapered buffer chain, and its power budget explicitly accounts for the clock distribution network, which is precisely the overhead that the asynchronous design aims to eliminate. The comparison between the two converters therefore isolates the effect of removing the global clock from an otherwise structurally similar Flash architecture.

A critical practical challenge of the slicer-based thermometer coder is its sensitivity to process variation. Since the threshold of each inverter is determined by the ratio of PMOS to NMOS drive strength, any departure from the nominal device parameters shifts the trip points in a bit-dependent manner, degrading the differential and integral non-linearity of the converter. To address this, an offline body bias trimming strategy was designed and implemented using a Verilog-A behavioural model. By applying distinct bulk voltages to the NMOS and PMOS transistors of the slicer stage, the threshold voltage of each device type can be shifted through the body effect, so the trip points can be steered back toward their nominal positions after fabrication. Correction vectors were determined for the four standard process corners (SS, FF, FS and SF) and were shown in simulation to improve the worst-case DNL at all four corners and the worst-case INL at three of them, with the largest gains at the symmetric corners; the SF corner proved the hardest to correct with a single global bias pair.

Finally, the complete system was validated through transistor-level analog and mixed-signal co-simulation. The dynamic performance of the asynchronous converter was characterised using a non-uniform to uniform resampling algorithm applied to the raw level-crossing output, enabling standard spectral metrics such as SNDR and ENOB to be extracted and compared against both

the synchronous reference and representative state-of-the-art designs. The power measurements confirmed the key theoretical prediction: dynamic power consumption scales linearly with input signal frequency, and the converter enters a near-zero dynamic consumption state when the input is idle. The adapted Walden figure of merit, computed using the effective average crossing rate rather than a fixed Nyquist frequency, demonstrates a clear energy efficiency advantage over the synchronous baseline for the sparse-signal operating regime targeted by this work.

The contribution of this work lies less in any single block than in their combination: ratioed-slicer quantisation, corner-indexed body bias trimming and clockless event generation, brought together at 0.8 V in a 16 nm process and benchmarked against a synchronous reference of equal resolution and technology.

7.2 Concluded Objectives

The four specific objectives defined at the outset of this dissertation were all successfully accomplished.

The first objective was the design of an asynchronous Flash ADC architecture that dissipates dynamic power only when input signal events occur. This was achieved through the final topology, a complementary five-transistor OTA driving the ratioed-slicer bank, which operates in continuous time without any global clock signal. The theoretical analysis confirmed that the converter's dynamic power scales linearly with the input signal frequency and approaches zero during idle intervals, directly validating the intended activity-dependent power profile.

The second objective was the implementation of an offline trimming mechanism capable of compensating the deterministic trip-point deviations of the comparison stage across process corners. This was fulfilled through the body bias trimming strategy described in Chapter 4, in which pre-computed pairs of NMOS and PMOS bulk voltages are applied to the slicer bank to restore the intended trip-point positions. The approach was validated across the SS, FF, FS, and SF process corners, reducing the worst-case DNL at all corners and the worst-case INL at three of the four, as reported in Table 6.3.

The third objective was the validation of the complete system through analog and digital co-simulation. The transistor-level netlist of the asynchronous ADC was simulated in a 16 nm CMOS process under a 0.8 V supply, and both static linearity and dynamic performance metrics were extracted. The FFT-based spectral analysis, performed on the uniformly resampled level-crossing output, confirmed correct operation of the asynchronous core and the functionality of the trimming logic, satisfying this objective.

The fourth objective was performance benchmarking against synchronous state-of-the-art designs using established figures of merit. The proposed converter was compared against the synchronous reference Flash ADC implemented under identical technology and supply conditions, as well as against relevant designs from the ISSCC and JSSC literature. The adapted Walden figure of

merit confirms that the asynchronous architecture achieves superior energy efficiency for sparse-signal operating conditions, closing a meaningful portion of the power gap that synchronous Flash converters exhibit at high sampling rates.

It should be noted that, although process corner variation was successfully addressed by the offline trimming strategy, Monte Carlo statistical variation was not corrected within the scope of this work. The body bias trimming vectors were determined for the four deterministic process corners and are therefore not adapted to the continuous distribution of device mismatch that Monte Carlo analysis exposes. The Monte Carlo results of Chapter 6 quantify the cost: the spread of a single code width reaches several LSB. This limitation is acknowledged as an important gap between the presented prototype and a fully robust production-ready implementation, and it is the primary motivation for the first suggestion for future work described in the following section.

7.3 Future Work

Two directions for future development are identified as the most impactful extensions of the work presented in this dissertation.

7.3.1 Analogue Process Corner Detection Circuit

The trimming strategy validated in this work relies on the correct identification of the active process corner, which in the current implementation is supplied externally to the Verilog-A behavioural model and used to select the appropriate pre-computed correction vector. In a physical integrated circuit this external identification is not available: the fabricated die has no direct means of knowing which combination of NMOS and PMOS device parameters it has been assigned by the foundry.

A natural extension would therefore be a dedicated on-chip analogue process detection circuit that senses the effective transistor characteristics of the fabricated device and communicates the inferred process corner directly to the trimming logic. Such a circuit could be implemented as a small reference cell: for example, a pair of matched inverter-based ring oscillators, one biased toward NMOS-dominated operation and the other toward PMOS-dominated operation, whose relative oscillation frequencies or propagation delays encode the current process corner. The ratio between the two delay measurements maps unambiguously onto the NMOS/PMOS balance of the process and distinguishes SS from FF and FS from SF corners. The resulting analogue measurement would be digitised by a simple comparator and used to address a lookup table of pre-computed body bias correction values, replacing the external corner identification currently provided by the Verilog-A model. This approach would make the trimming procedure fully self-contained and autonomous, removing the need for any external process characterisation while preserving the static, low-overhead nature of the offline correction strategy.

7.3.2 Monte Carlo Trimming

A more fundamental limitation of the current trimming approach is that it corrects deterministic inter-corner shifts but leaves intra-die stochastic mismatch unaddressed. In practice, even at the nominal process corner, individual transistors within the slicer bank will exhibit random threshold voltage variations whose statistical distribution is characterised by Pelgrom's law [104]. These variations are not captured by the four-corner trimming table, and their cumulative effect on DNL and INL can be significant in a Flash converter where each comparator threshold must be individually accurate.

Extending the calibration to handle Monte Carlo variation would require moving from a corner-indexed lookup table to a converter-specific foreground calibration routine capable of measuring the actual trip point of each slicer independently and computing an individual correction for each one. One viable approach would be to inject a known ramp stimulus at the converter input during a dedicated startup calibration phase and record the digital code transitions to infer the actual trip point of each slicer. The deviation of each measured trip point from its nominal target could then be mapped to the body bias correction required to bring it back into specification, using the sensitivity relationship established in Chapter 4. Since the NMOS and PMOS bulk voltages of each slicer would need to be individually controlled, this approach requires the addition of a per-cell bias generator, which increases the area overhead relative to a global corner correction. Nevertheless, the achievable linearity improvement and the corresponding reduction in post-fabrication yield loss would likely justify this added complexity in any application where the converter is produced at volume and must meet tight DNL specifications across the full statistical distribution of process variability.

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